

HOWARD

TIMLIN

REPRESENTATIONS

$$\rho : G \rightarrow GL(V)$$

EDUCATION

Harvard Graduate School of Design
Masters of Architecture I

Expected May 2028

Harvard University
AB Mathematics & Physics

May 2023
3.93 GPA

ARCHITECTURE WORK

Developable Tectonic Research and Fabrication
June 2025 - August 2025

project:if
Boston, MA

- Team developed novel fabrication technique for developable geometries using standard-sized 4×8' bendy plywood.
- Constructed and installed exhibit, **Thin Volumes: In the Round**, in the 2025 Chicago Architecture Biennial.

Carbon Form Research
June 2025 - August 2025

Outside Development
Boston, MA

- Conducted historical and speculative drawing research for publication in development.

Computational Designer
November 2023 - August 2024

Certain Measures
(Hybrid) Boston, MA

- Lead front end developer on interactive environmental simulation application for the Wright-Ingraham Institute.
- Developed quality assurance system on custom CAD software for mass customized medical device.

AR Architecture Intern
September 2023 - November 2023

wa.k studio
Brooklyn, NY

- Designed and implemented interactive augmented reality applications for architectural models.

Technical TA for Collaborative Design Engineering Studio
September 2022 - October 2022

Harvard GSD
(Remote) Cambridge, MA

- Guided students of varying technical backgrounds through the process of using code and machine learning to create design engineering product prototypes.

3D Scanning & NeRF Interpolation Research
May 2022 - August 2022

GSDAI - Harvard GSD
(Hybrid) Cambridge, MA

- Implemented software for generating neural radiance fields (NeRFs) and 3D meshes with accurate material & lighting properties from cell phone videos.
- Created automated scanning protocol for robot arm using Rhino/Grasshopper interface.
- Created method for generating interpolations between NeRF representations.

Morphological Metrics for Generative Design Research
May 2021 - August 2021

GSDAI - Harvard GSD
(Remote) Cambridge, MA

- Leveraged techniques from geometry and topology to organize and simplify latent space outputted by 3D shape generative adversarial network (GAN).
- Developed automated pipeline for automatically generating and classifying 3D shape data.

PHYSICS RESEARCH

Nonlinear Duality in Mechanical Metamaterials and Chemical Reaction Networks (CRNs)

May 2020 - May 2021

Vitelli Group - University of Chicago

(Remote) Chicago, IL

__ Studied symmetries between different configurations of dynamical systems (called 'dualities') which had been shown to induce hidden full symmetries and unique properties in mechanical metamaterials.

__ Showed existence of nonlinear dualities in chemical reaction networks and reaction diffusion systems.

__ Developed Python simulation of reaction diffusion systems to test dualities.

Modeling and Controlling Branched Flow

May 2019 - December 2019

Heller Group - Harvard University

Cambridge, MA

__ Worked on computing cluster application in C++ to simulate wave propagation via the split-operator approach for solving the non-linear Schrödinger equation.

Enceladus Plume Modeling

August 2017 - August 2018

Southwest Research Institute (SwRI)

San Antonio, TX

__ Investigated novel, tidal model for crack compression responsible for producing geysers on Saturn's moon, Enceladus.

__ Created scale model of Enceladus crack conditions under vacuum to test theoretical predictions.

SKILLS

- __ Architectural Geometry
- __ Rhino 7/Grasshopper/V-Ray
- __ Architecture Model Fabrication
- __ Adobe Suite
- __ Microsoft Office
- __ Unity, C#, Javascript, HTML, Python, C++, Java
- __ Machine Learning

REPRESENTATIONS
of selected works
2021-2024

completed by
HOWARD TIMLIN

Candidate for the Master of Architecture I
Harvard University
Graduate School of Design
Expected May 2028

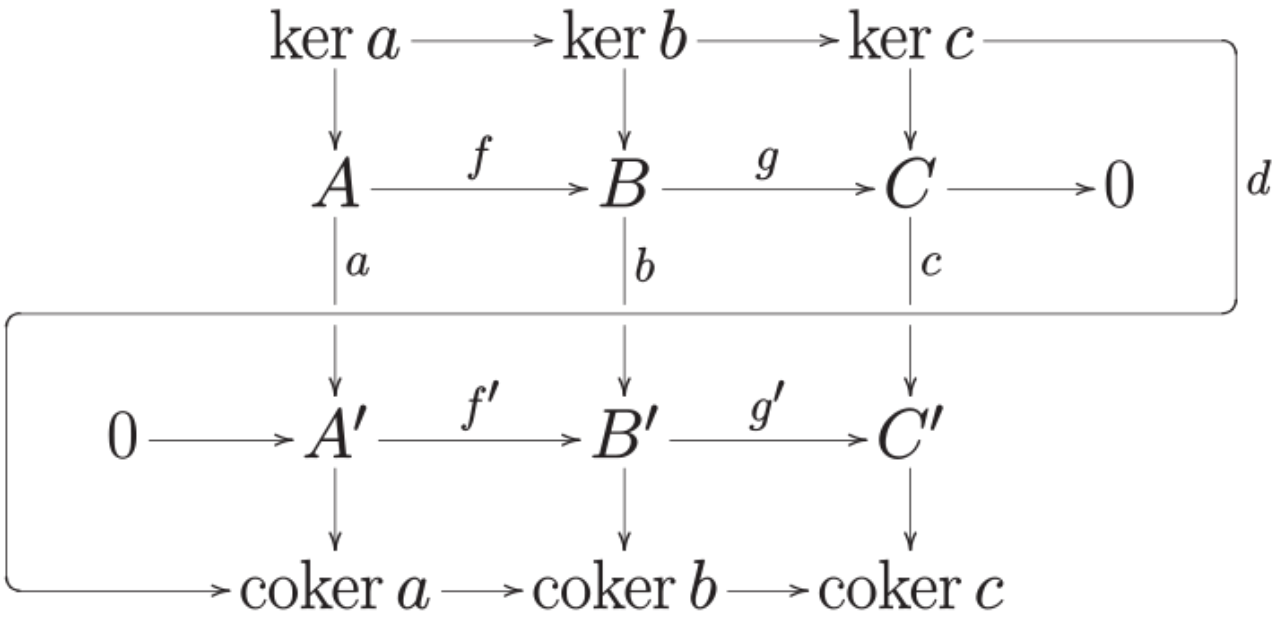
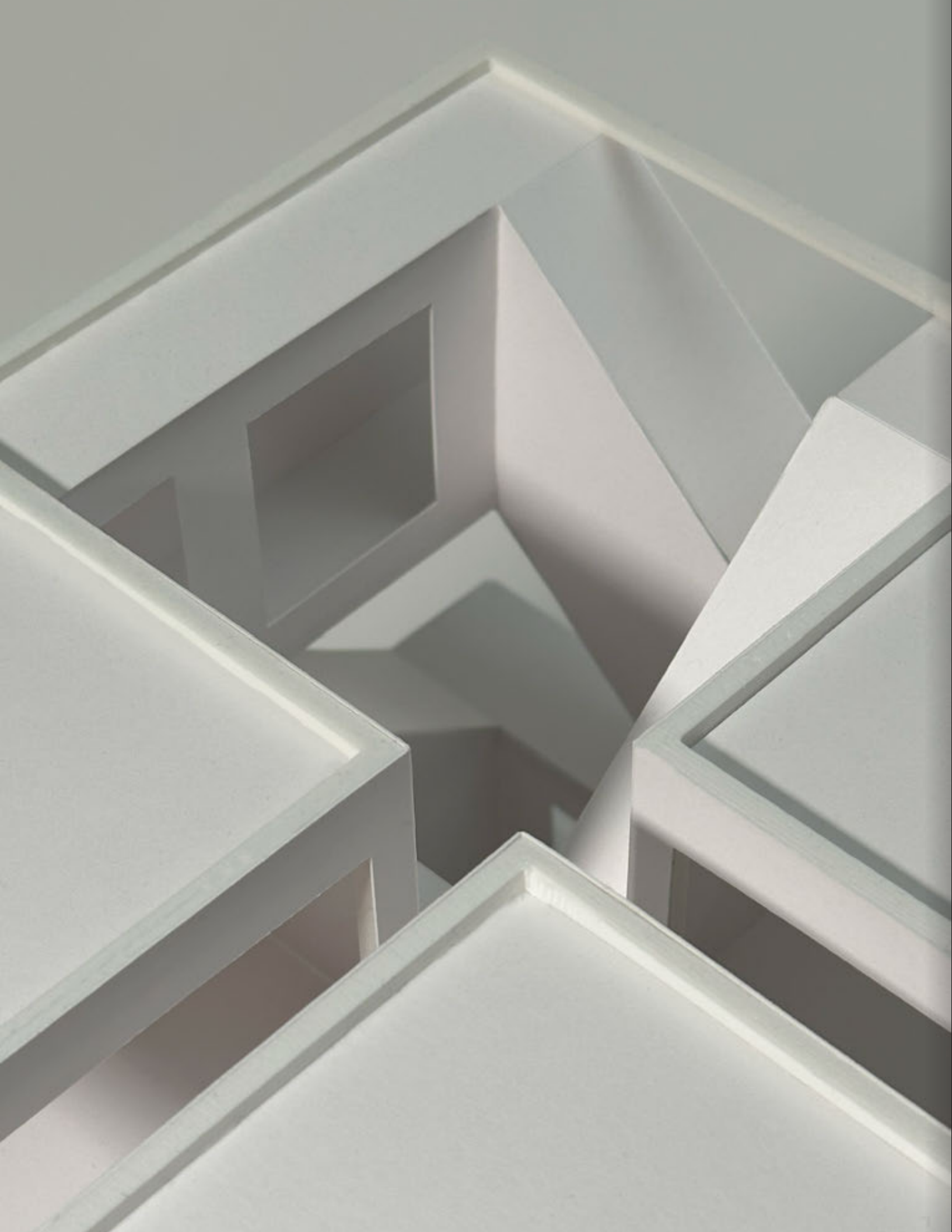
2	LIVING SINGLE LIVING TOGETHER
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CONTENTS

architecture

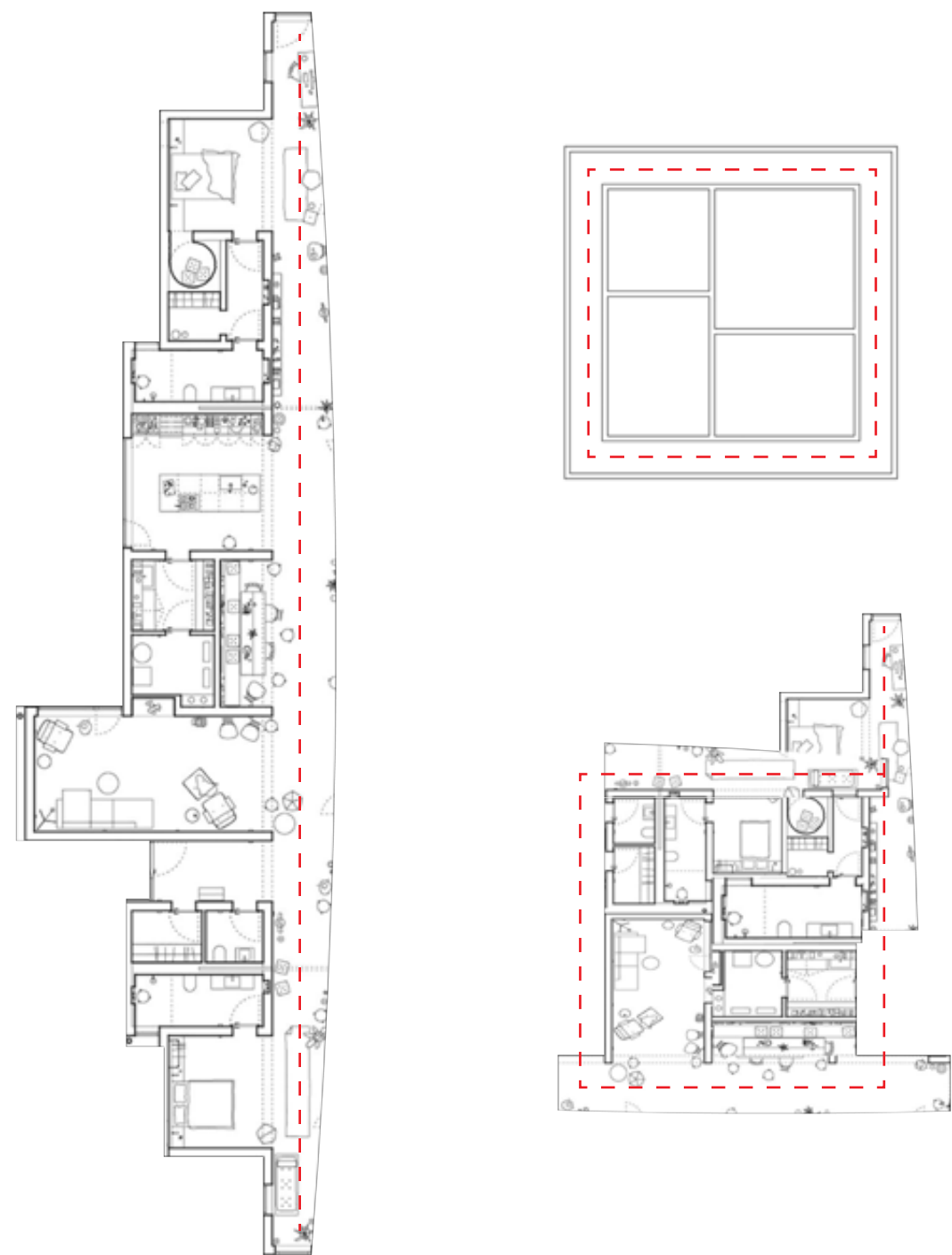


architecture?

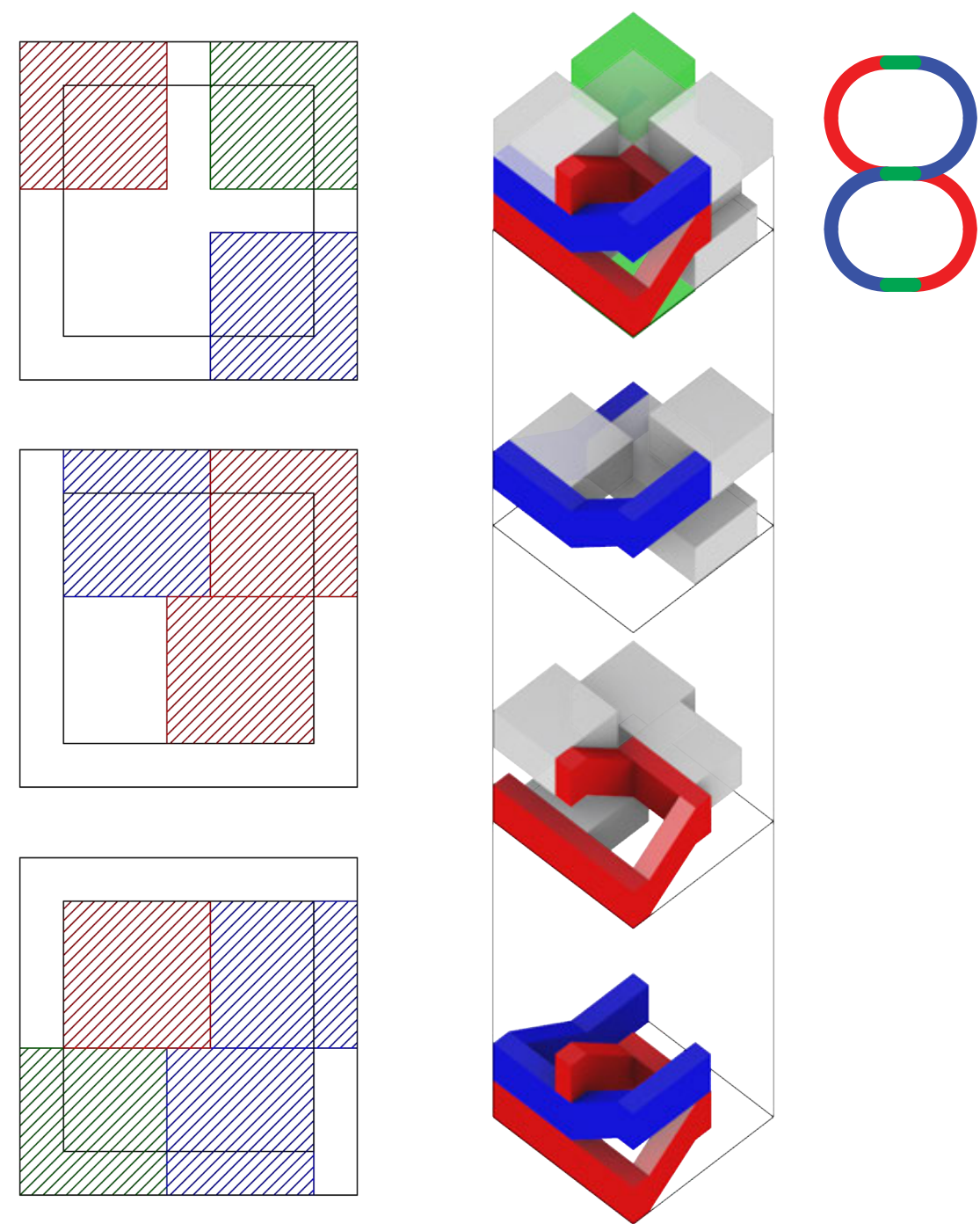


**LIVING SINGLE
LIVING TOGETHER**

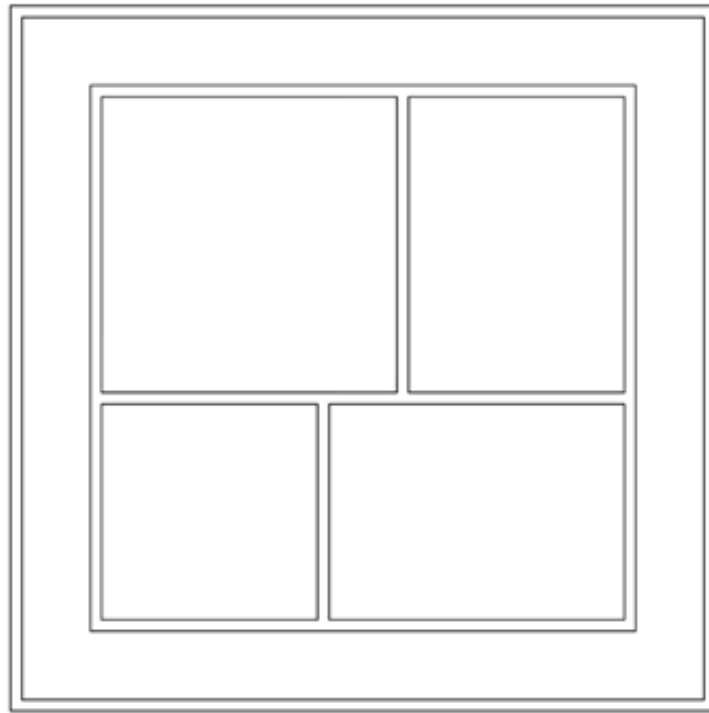
Brief: Extrapolate the given plan into a duplex accomidating two equally-sized residences with a mix of shared and private space.



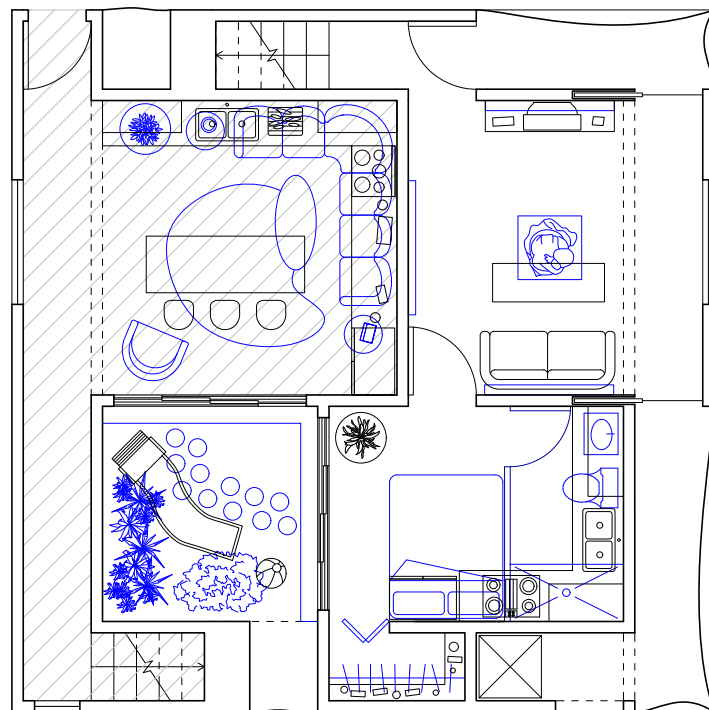
initial precedent study of **Hall House** by **WOJR**



reading of given plan as snapshot of spaces belonging to **Unit A**, **Unit B**, or **Shared Space** shifting to accomodate woven circulation

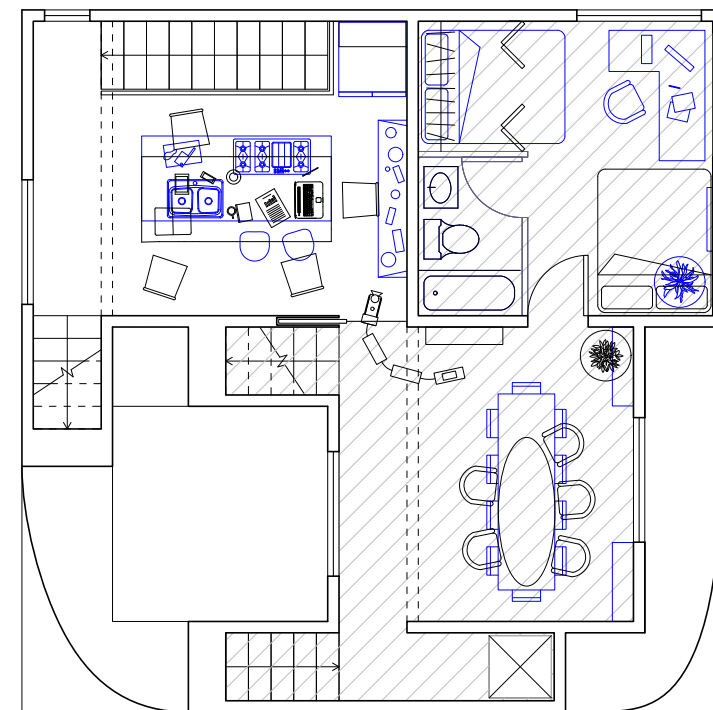
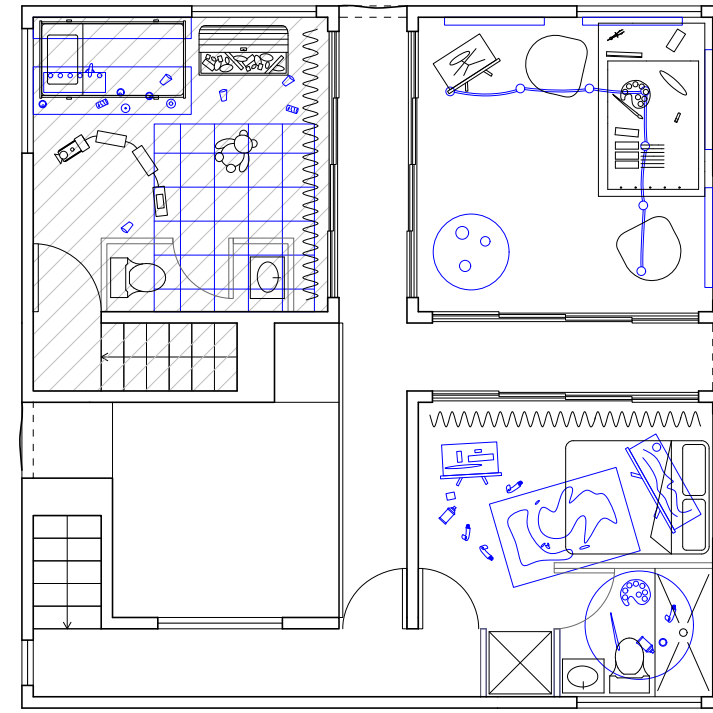


Unit A

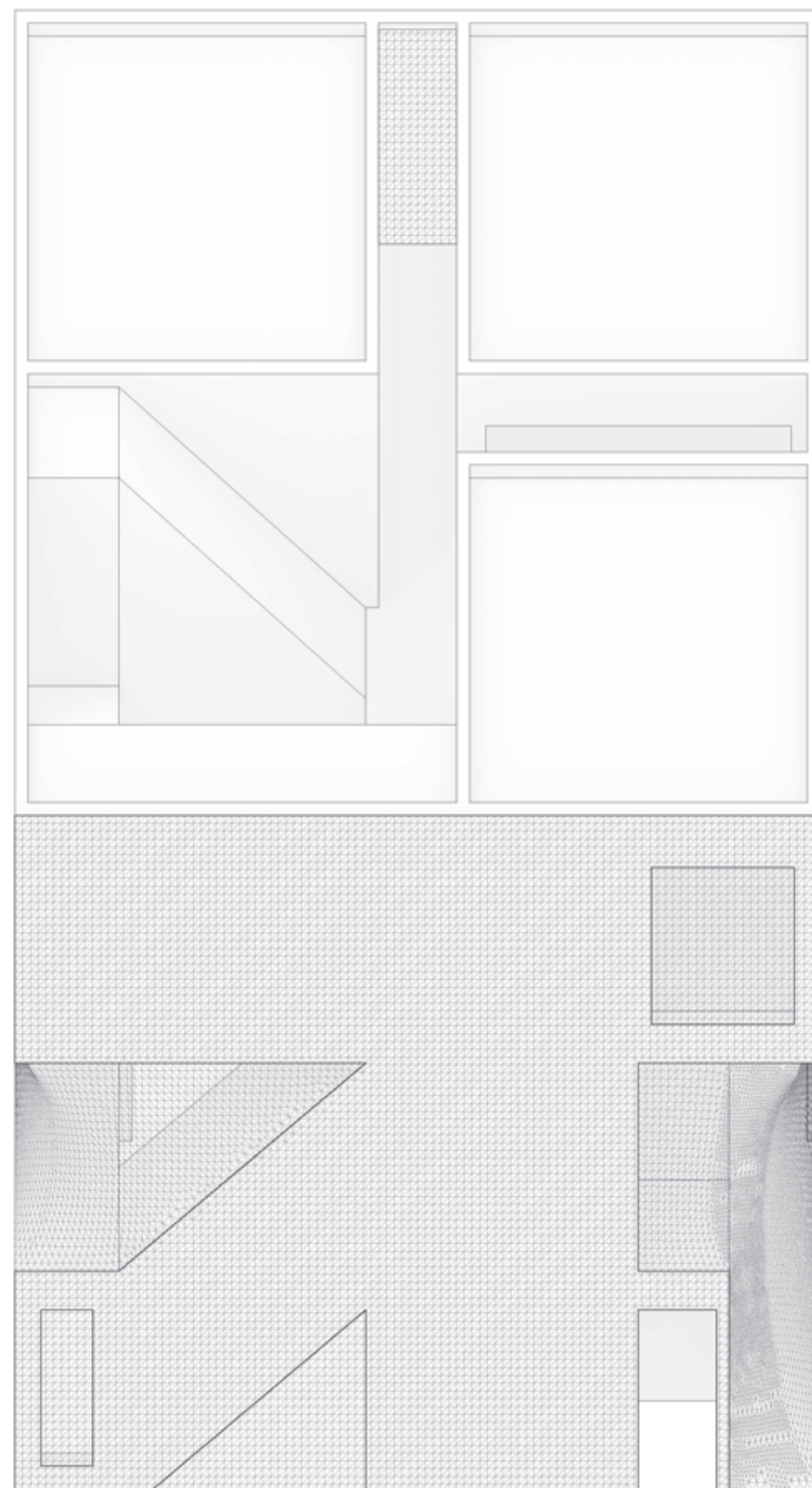
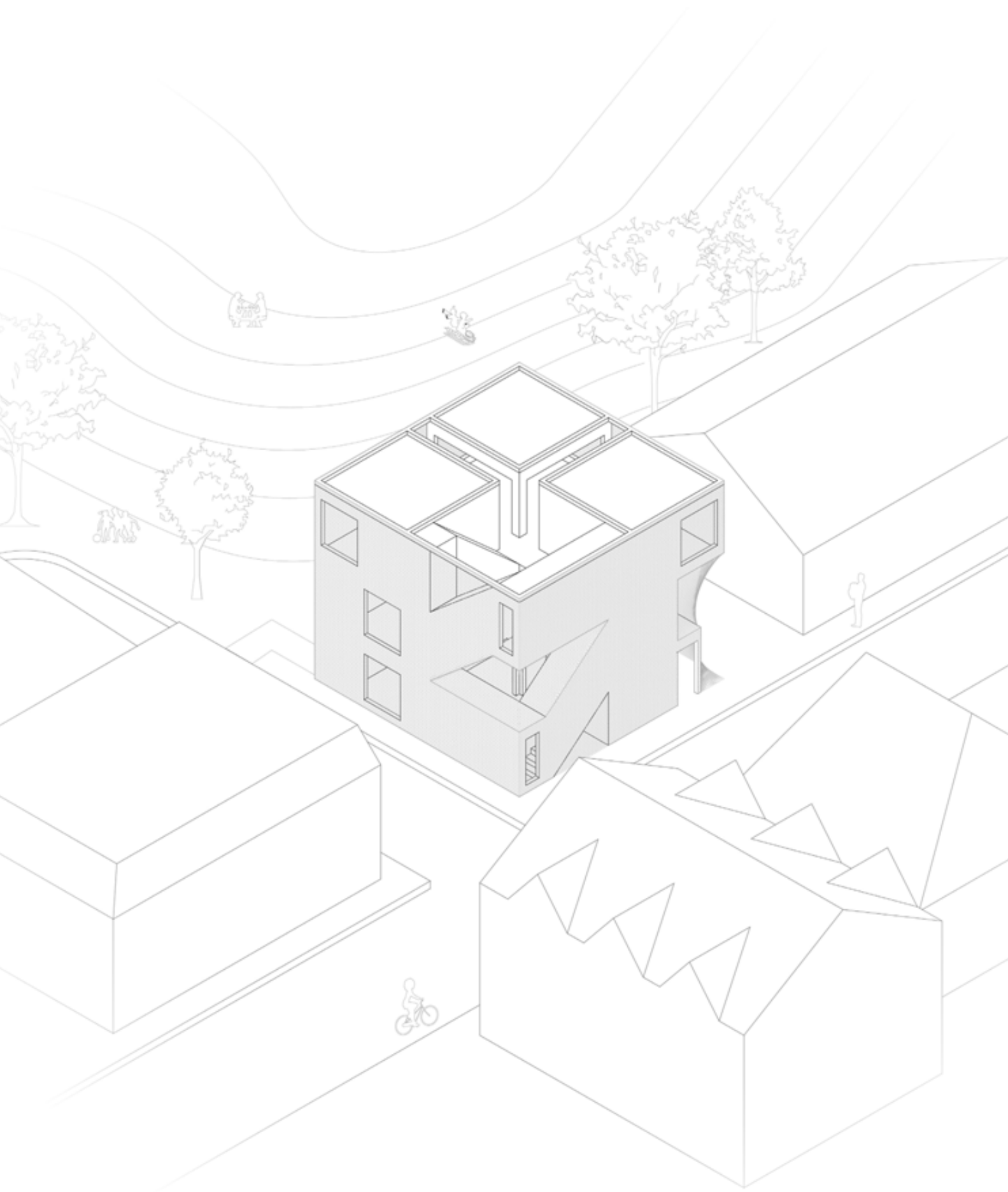


Unit B

resulting duplex accomodating multigenerational family, or...

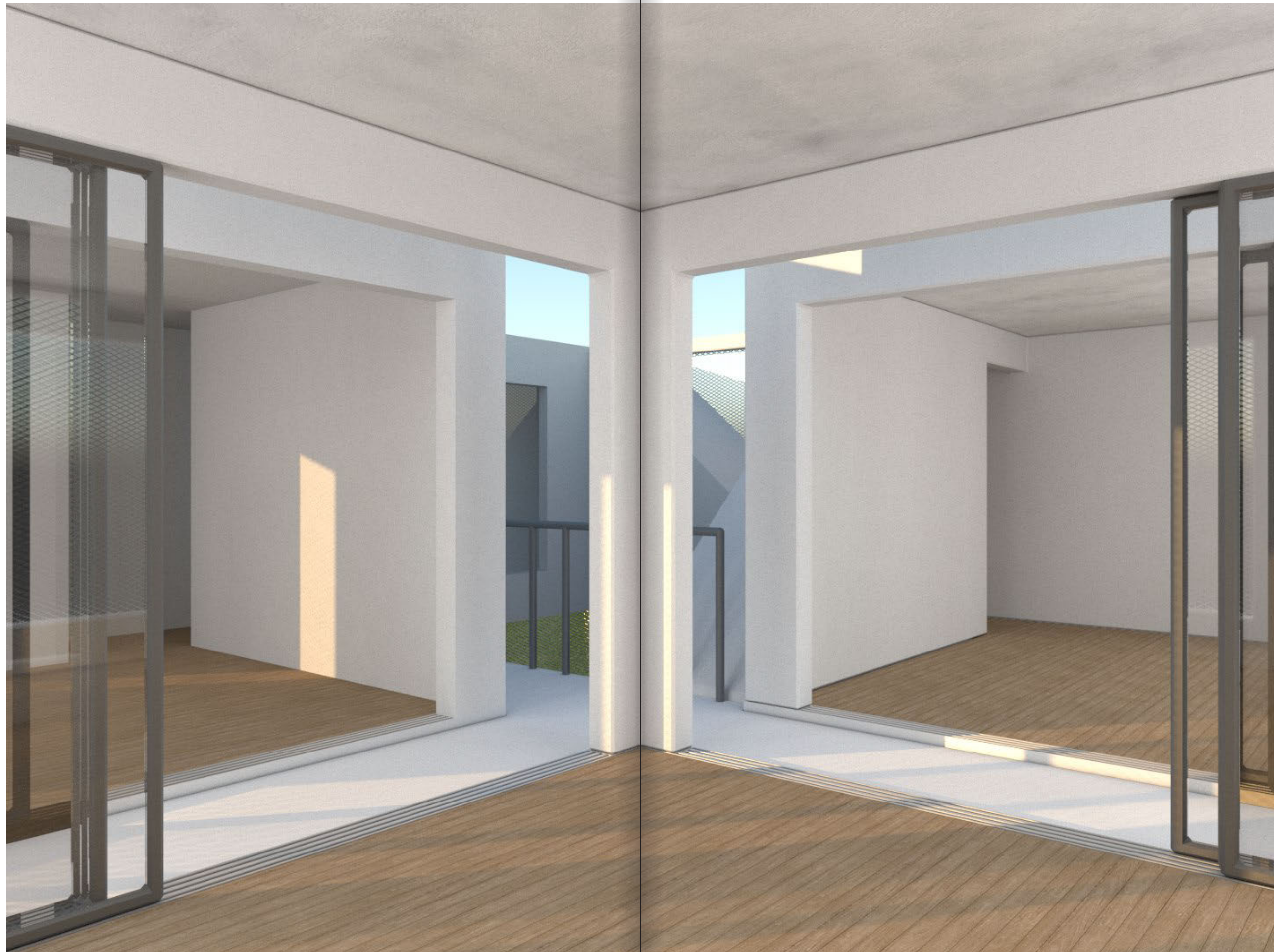


alternatively, an artist and wealthy benefactor





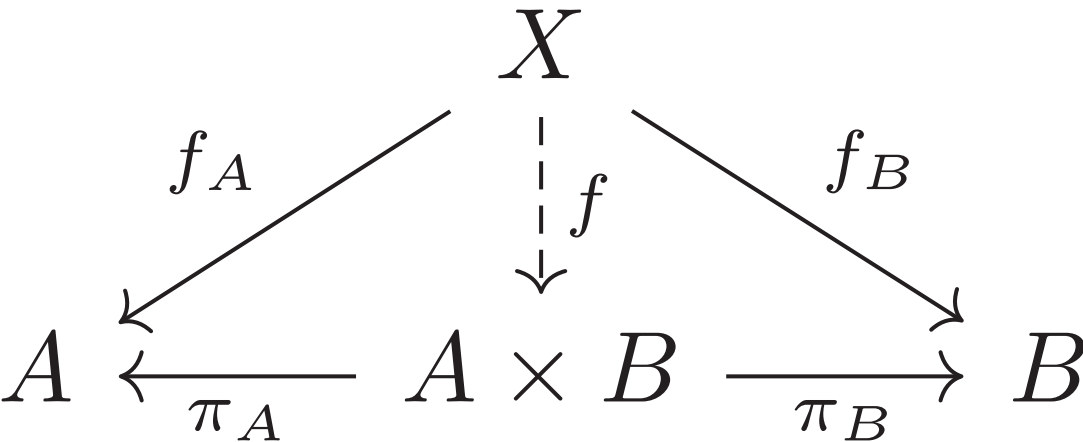






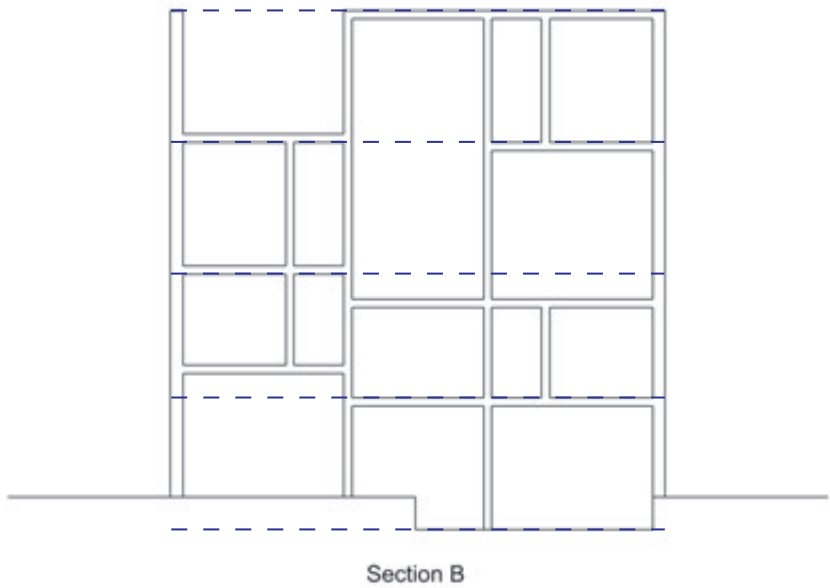
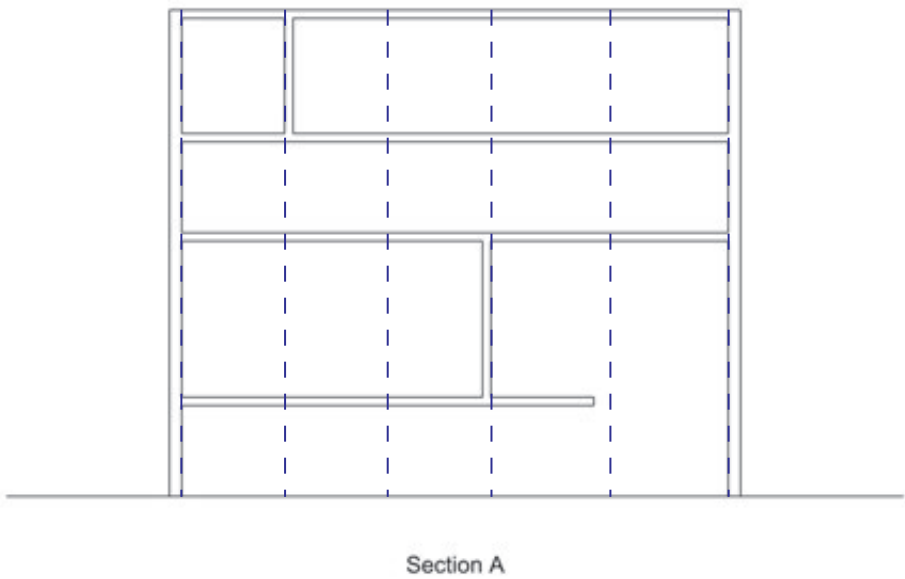
GSD STUDIO
CORE I

JENNY FRENCH
2024

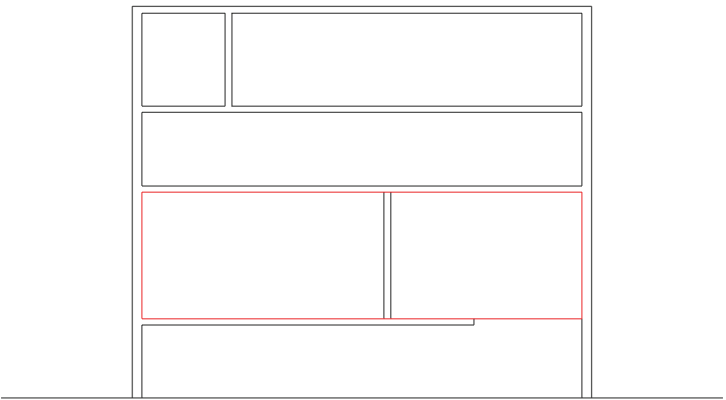


SAMPLE CUT STITCH

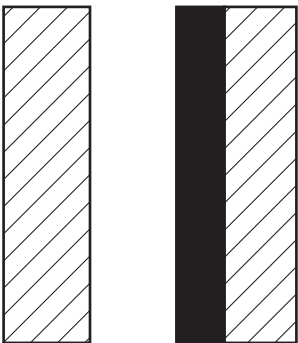
Brief: Reconcile two incongruous sections of a single building which is host to two disparate schools.



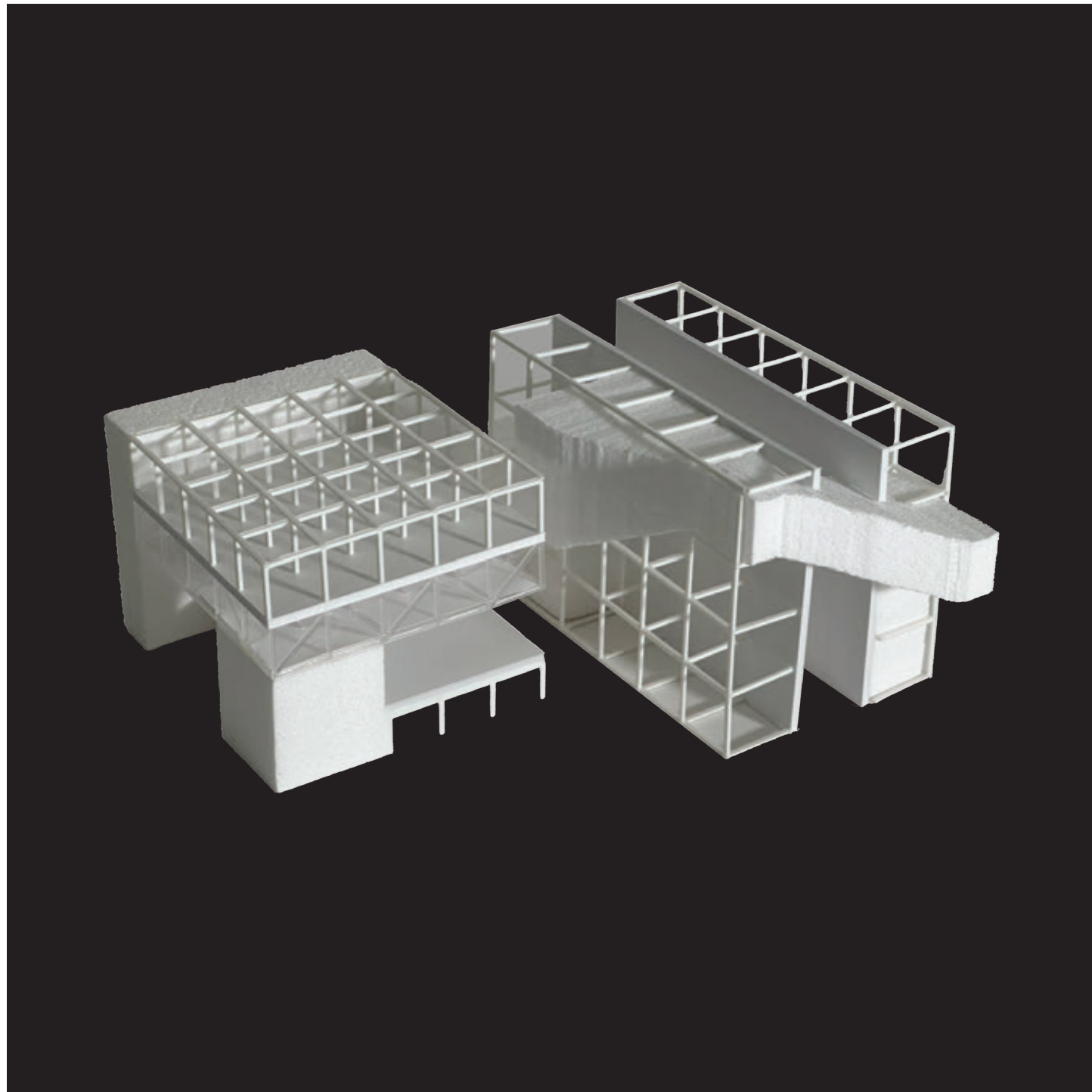
reading of structural grid within given sections



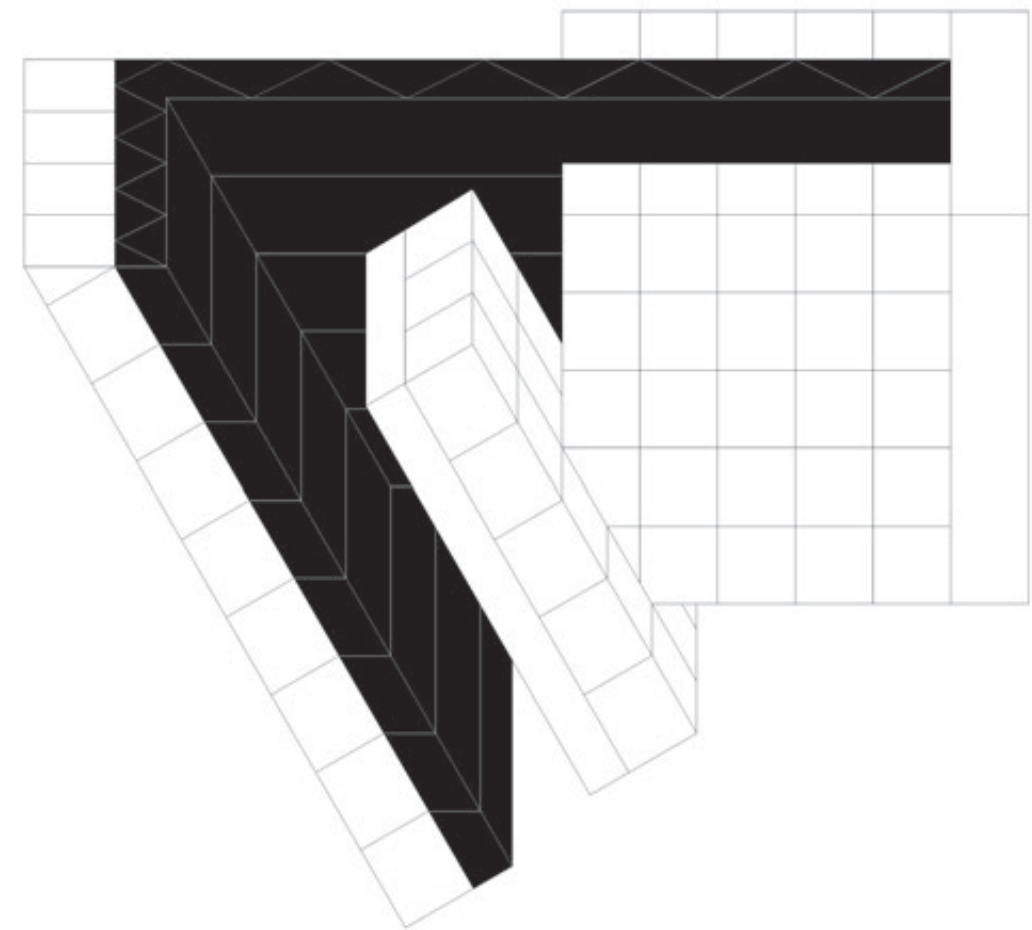
typological reading of assembly spaces: **studio** and **theater**



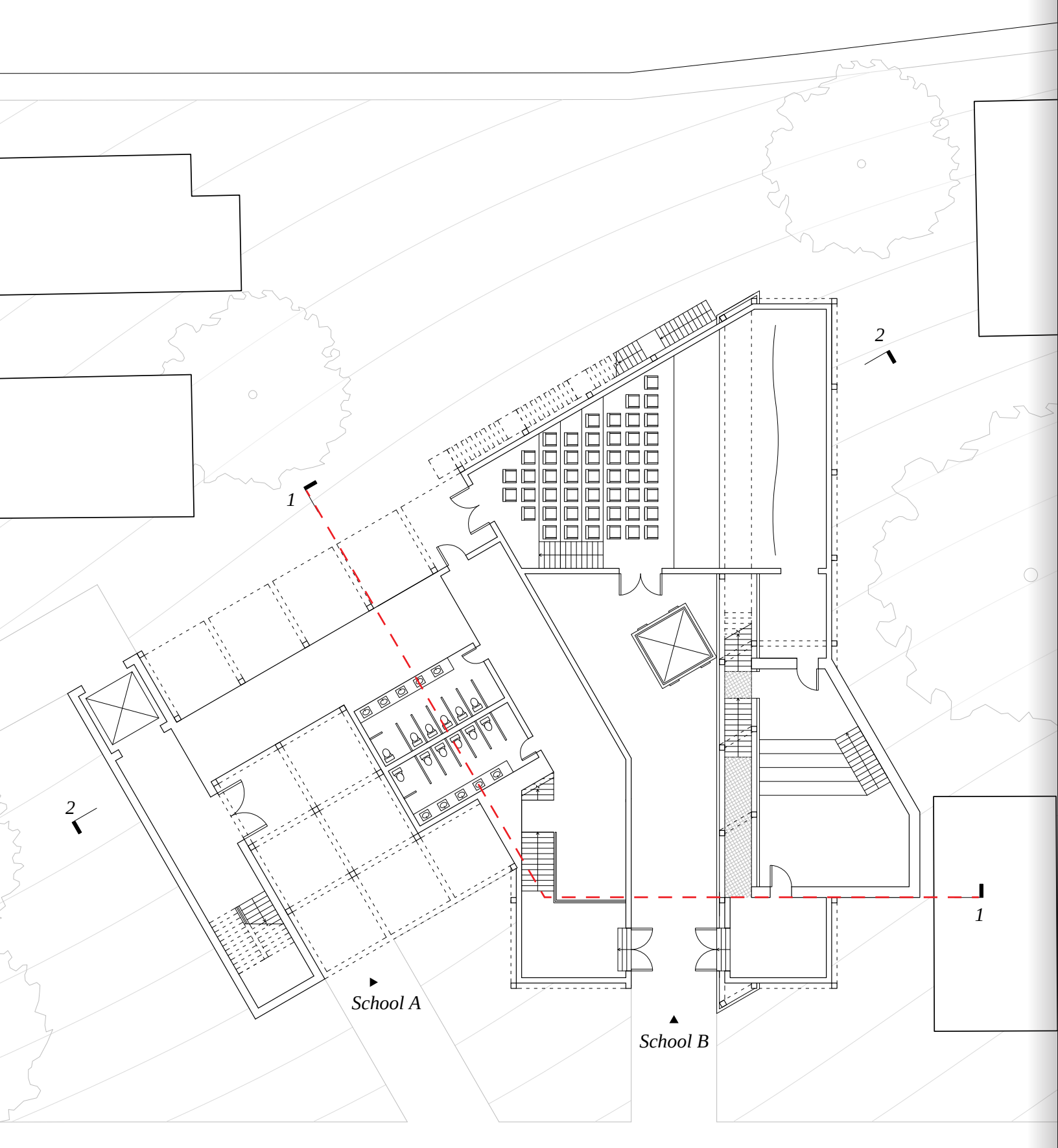
diagrammatic reading of sectional duality



extrapolation of read structural grids and typologies

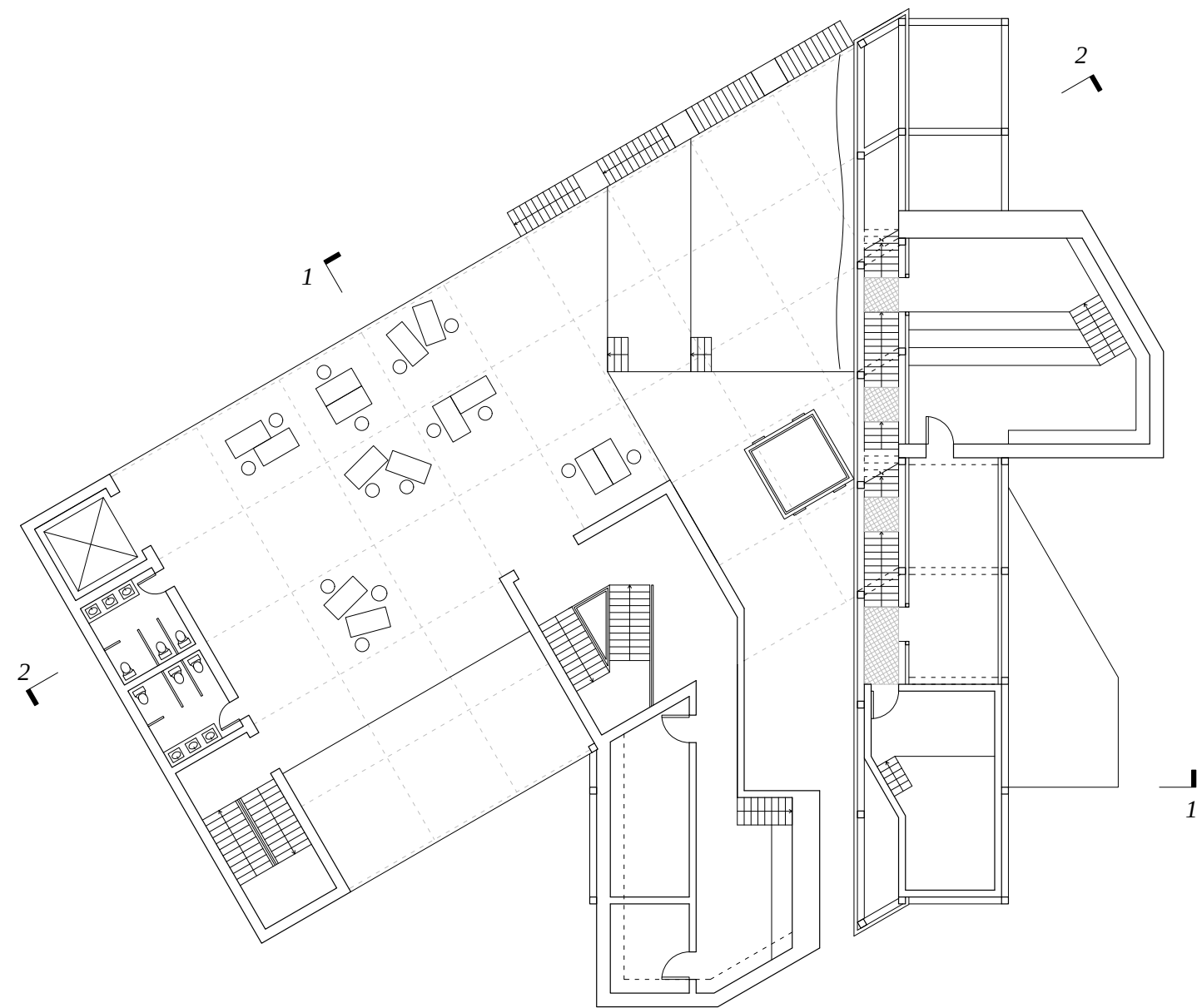


inhabitable truss forms common datum, mediating differences in typology and structural grid scale



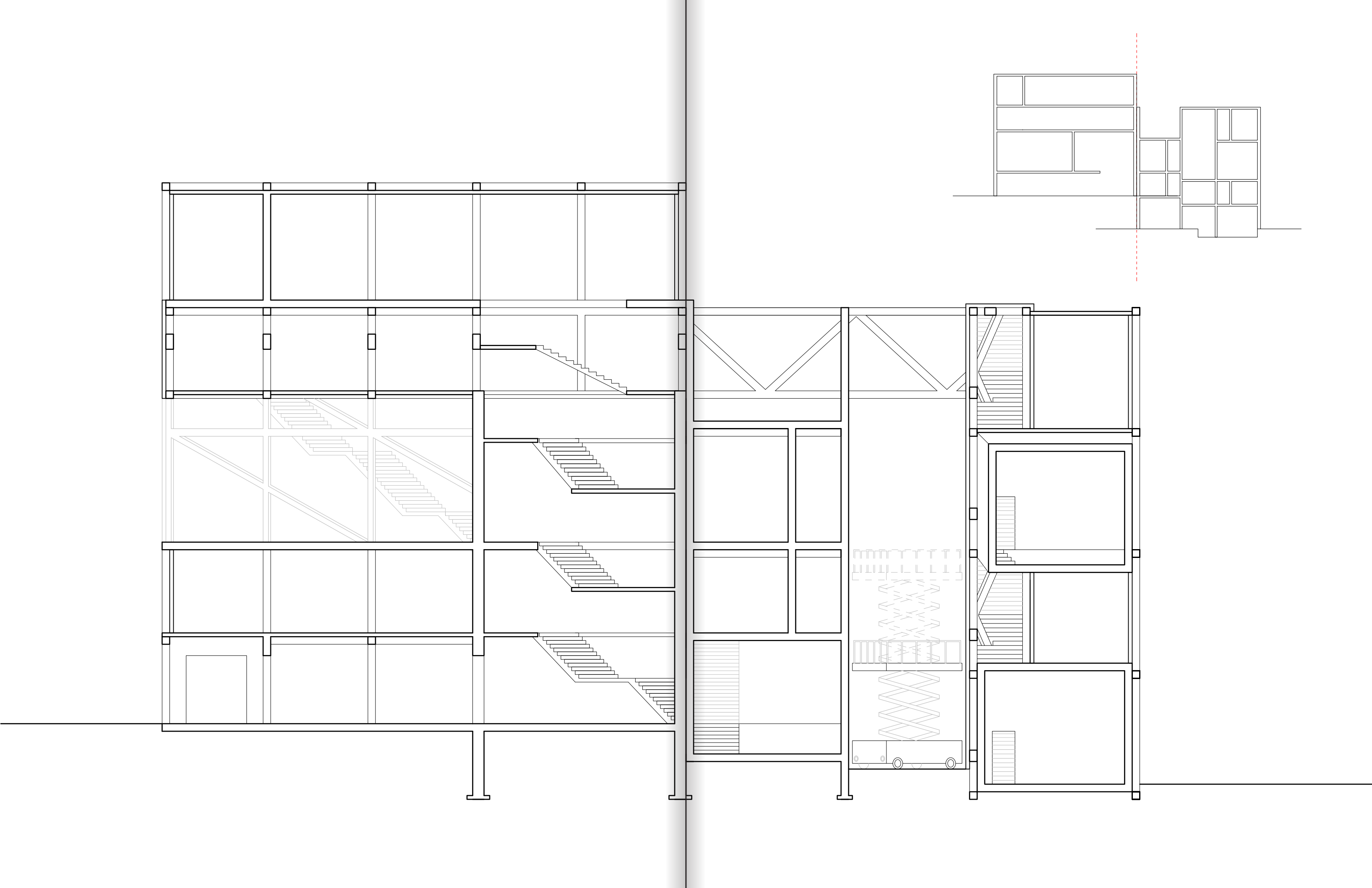
Plan 01

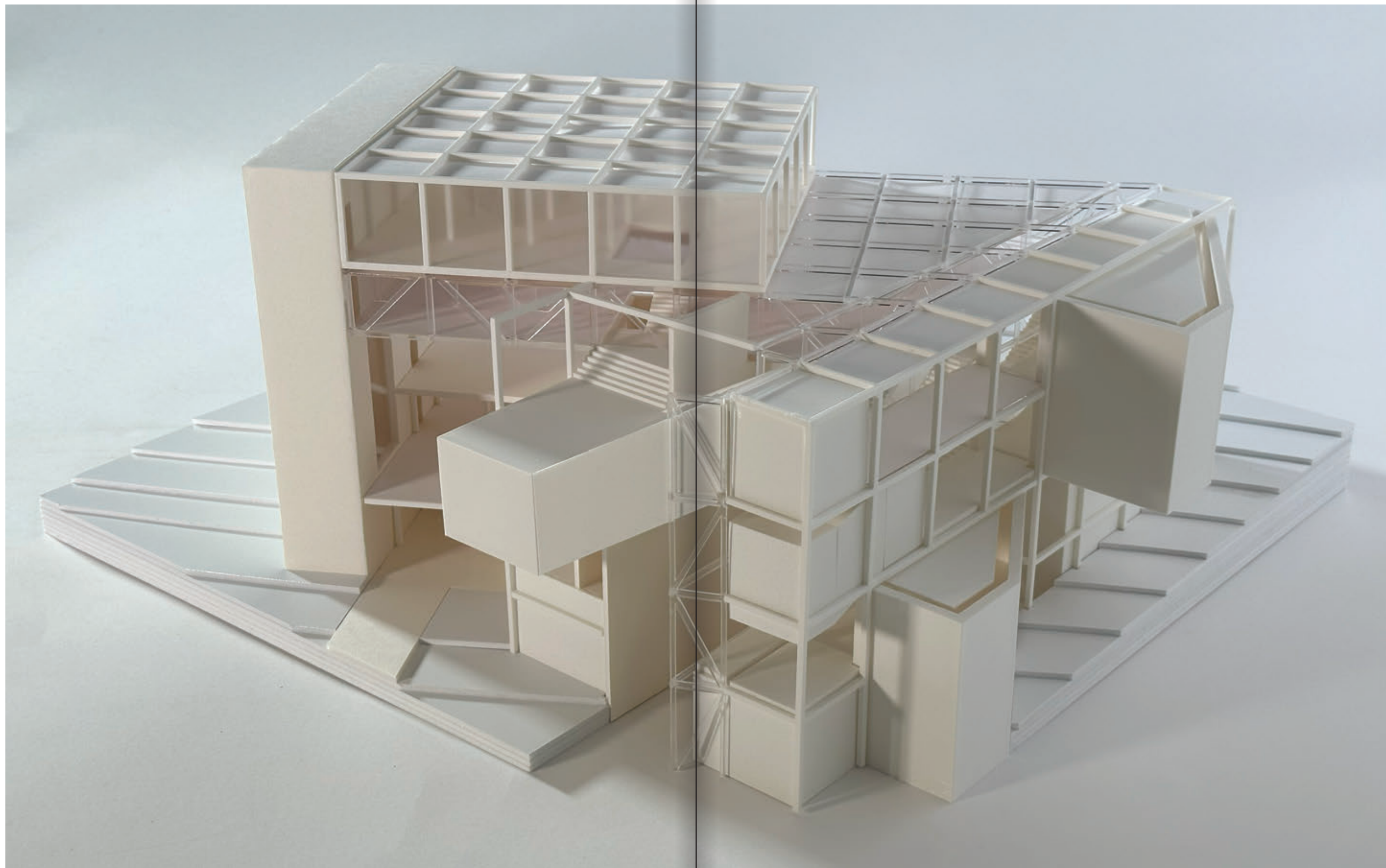
3/32" = 1'-0"



Plan 03

3/32" = 1'-0"

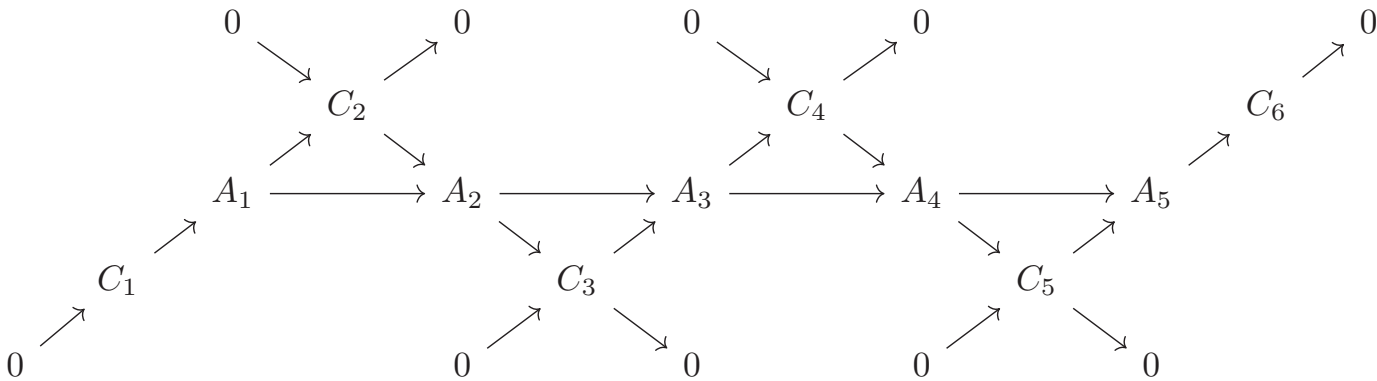
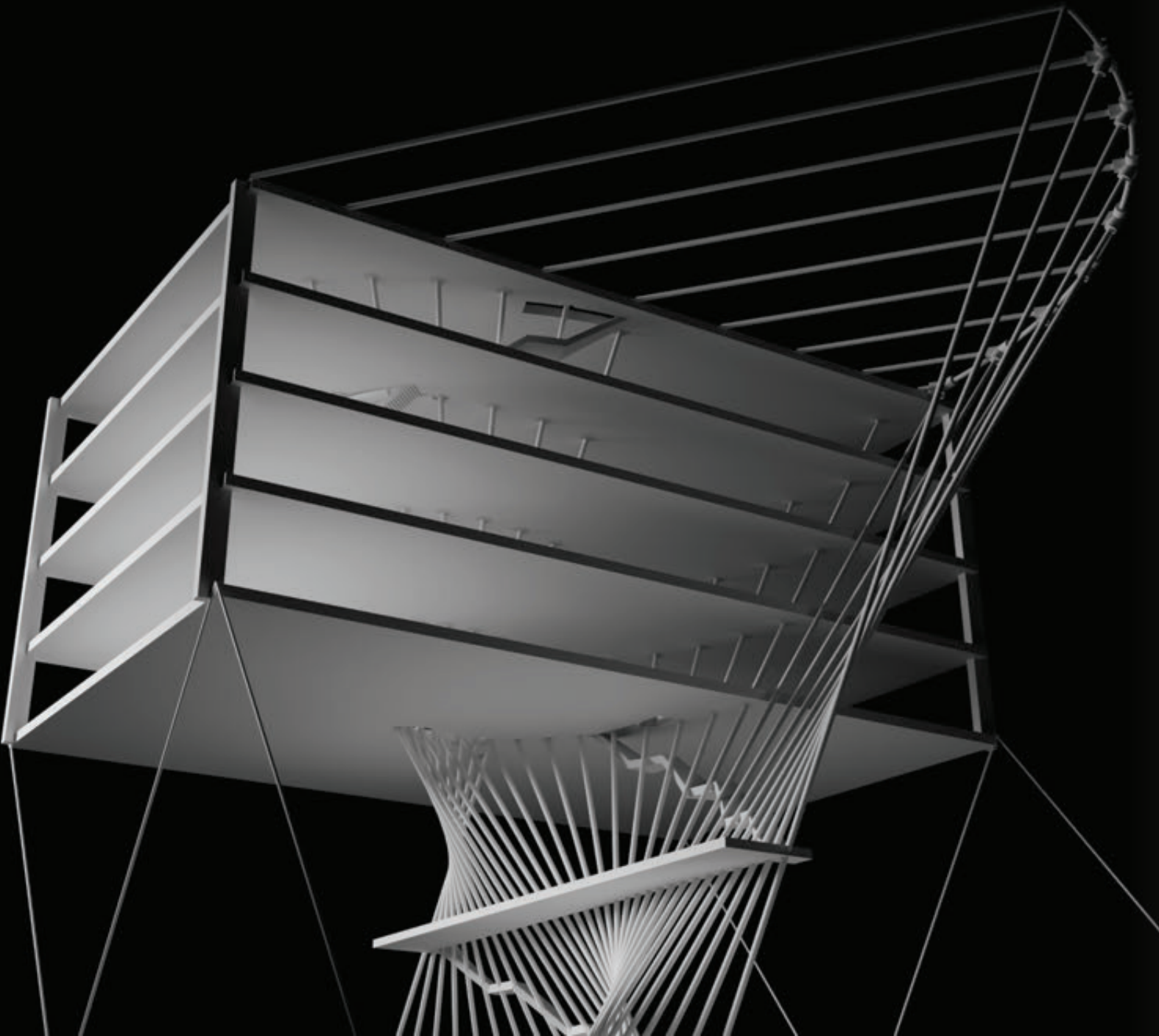




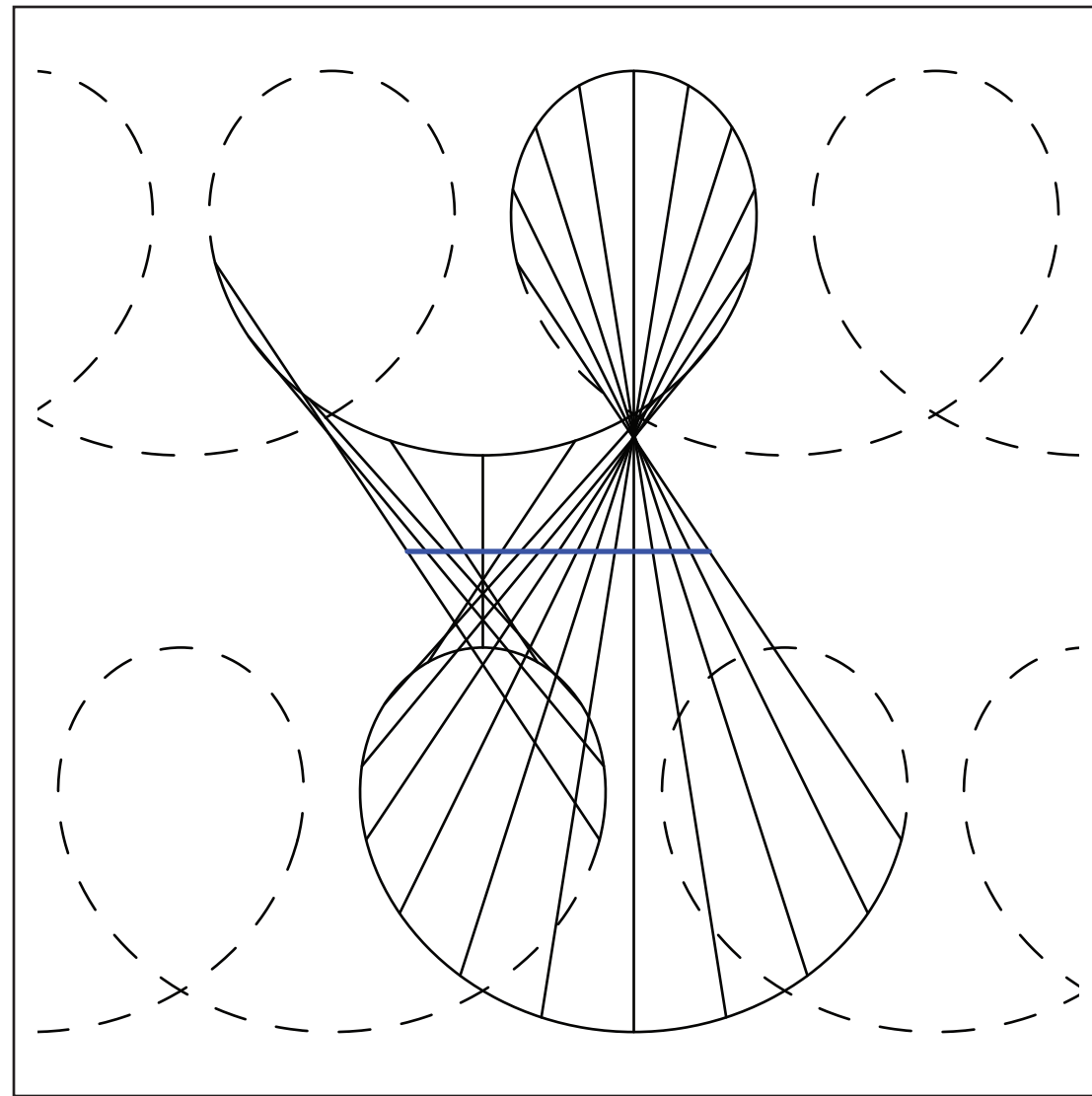
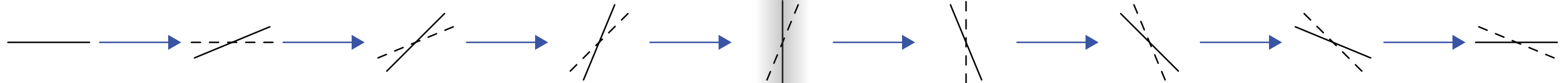


back of house + front of house: shared outdoor assembly space open to the community

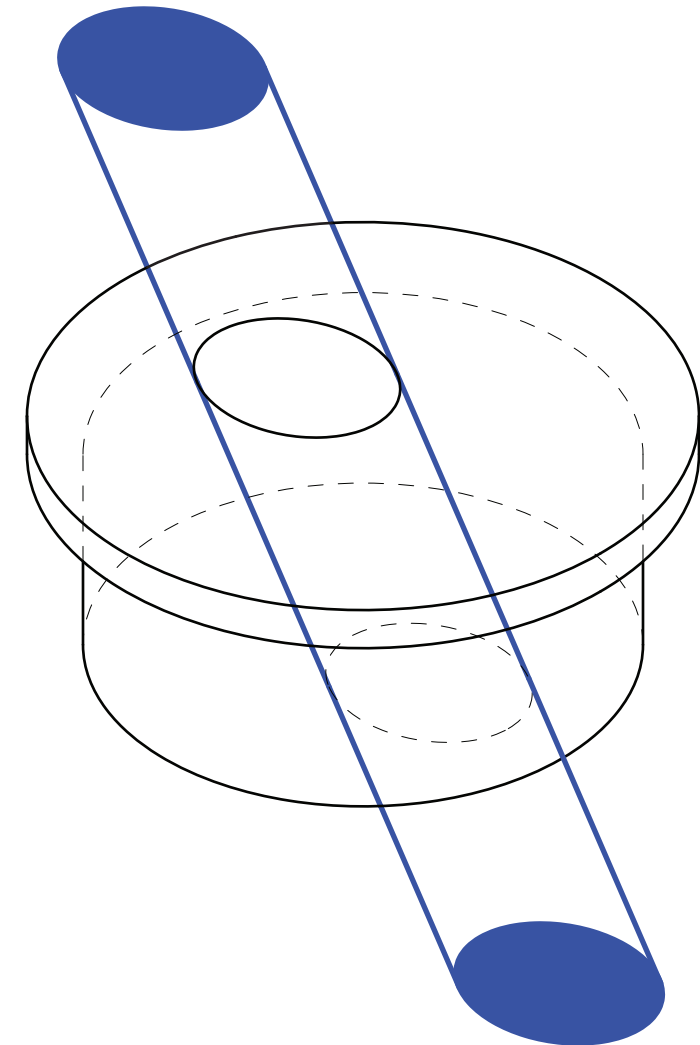




TROCHOID TOWER



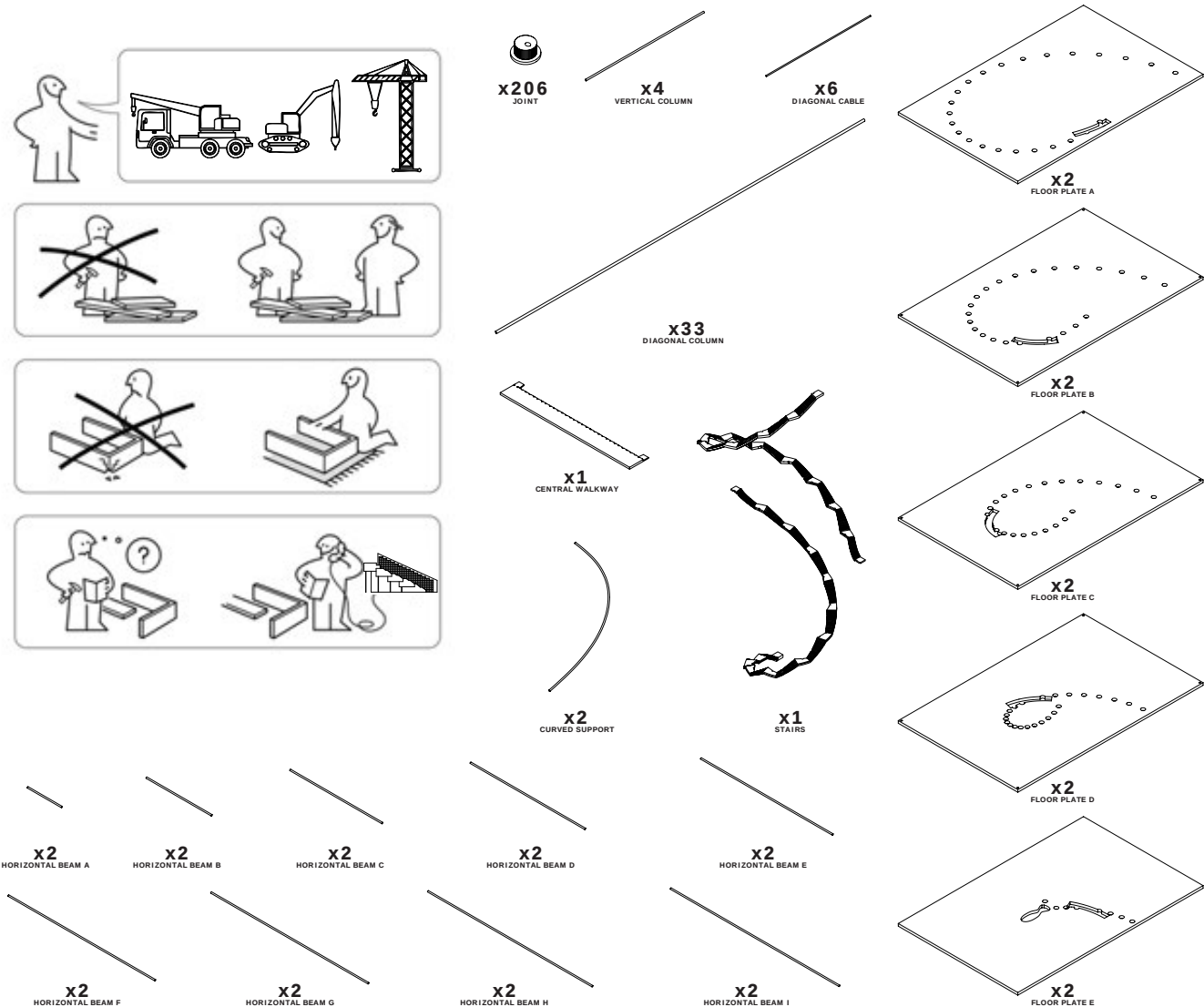
successive **translations** and **rotations** of an oblique column traces a prolate trochoid through space which passes through medial datum



the affine transformations of the oblique column preserve the relationship to floorplates, necessitating a **single type of joint**

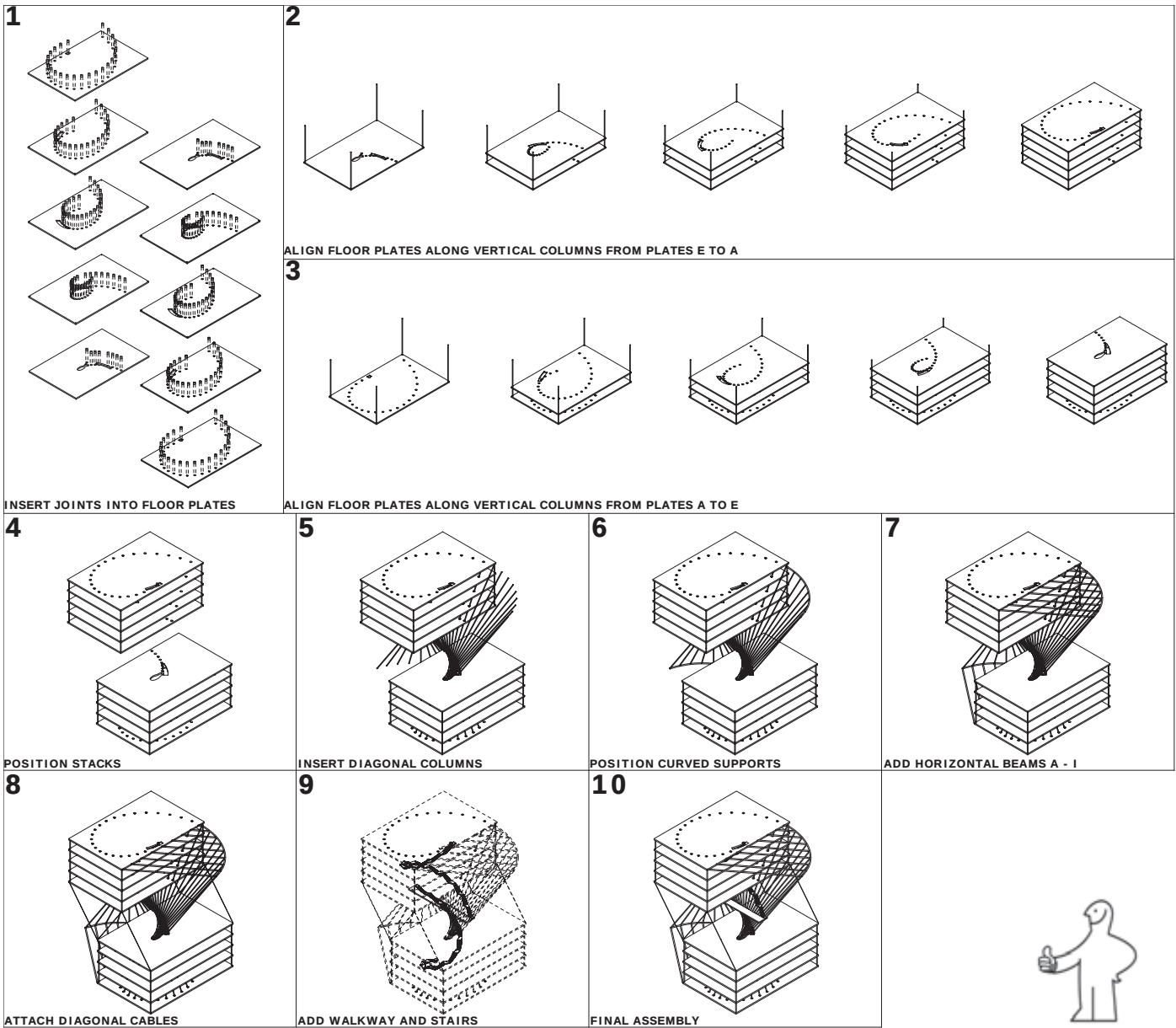
TROCHOID TOWER

KIT OF
PARTS

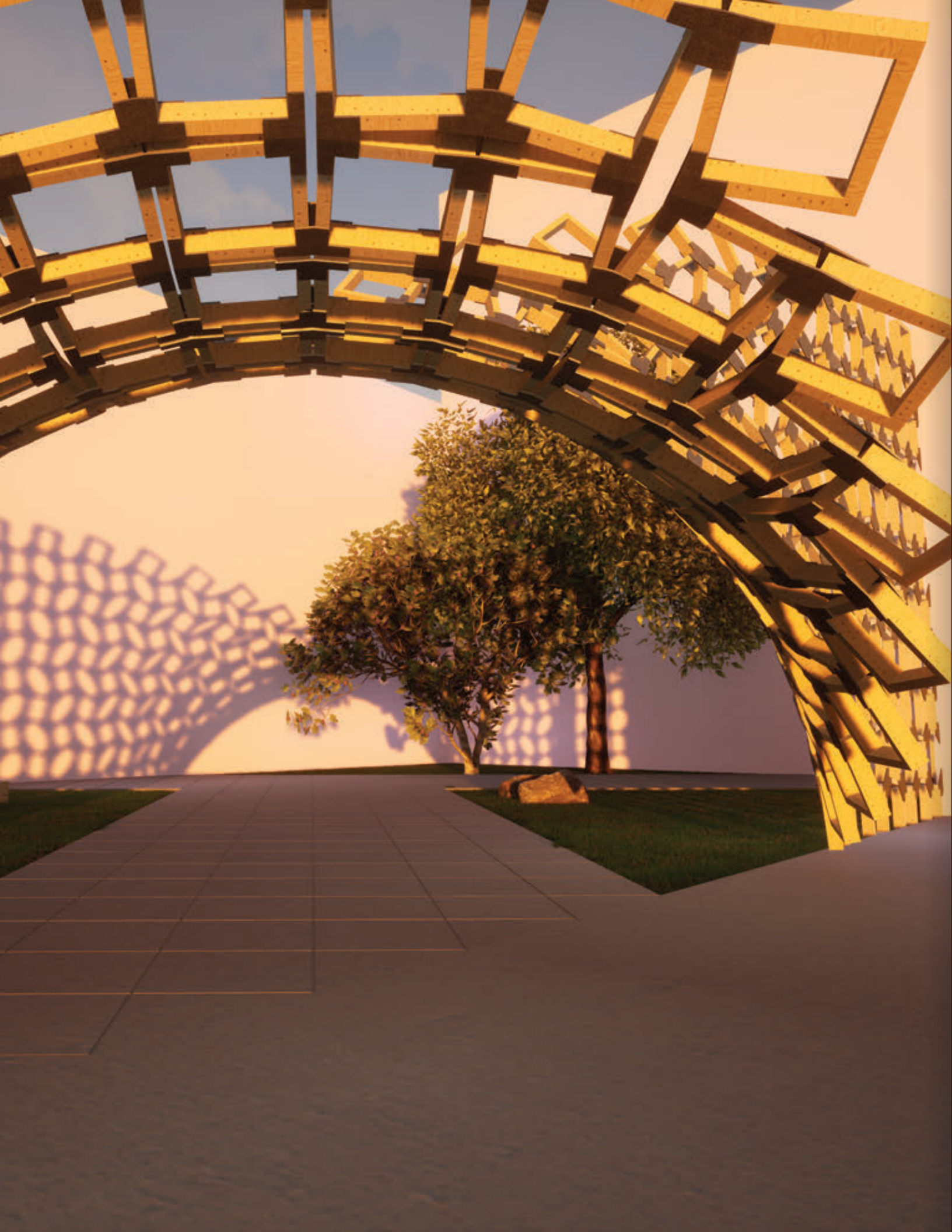


TROCHOID TOWER

ASSEMBLY
INSTRUCTIONS

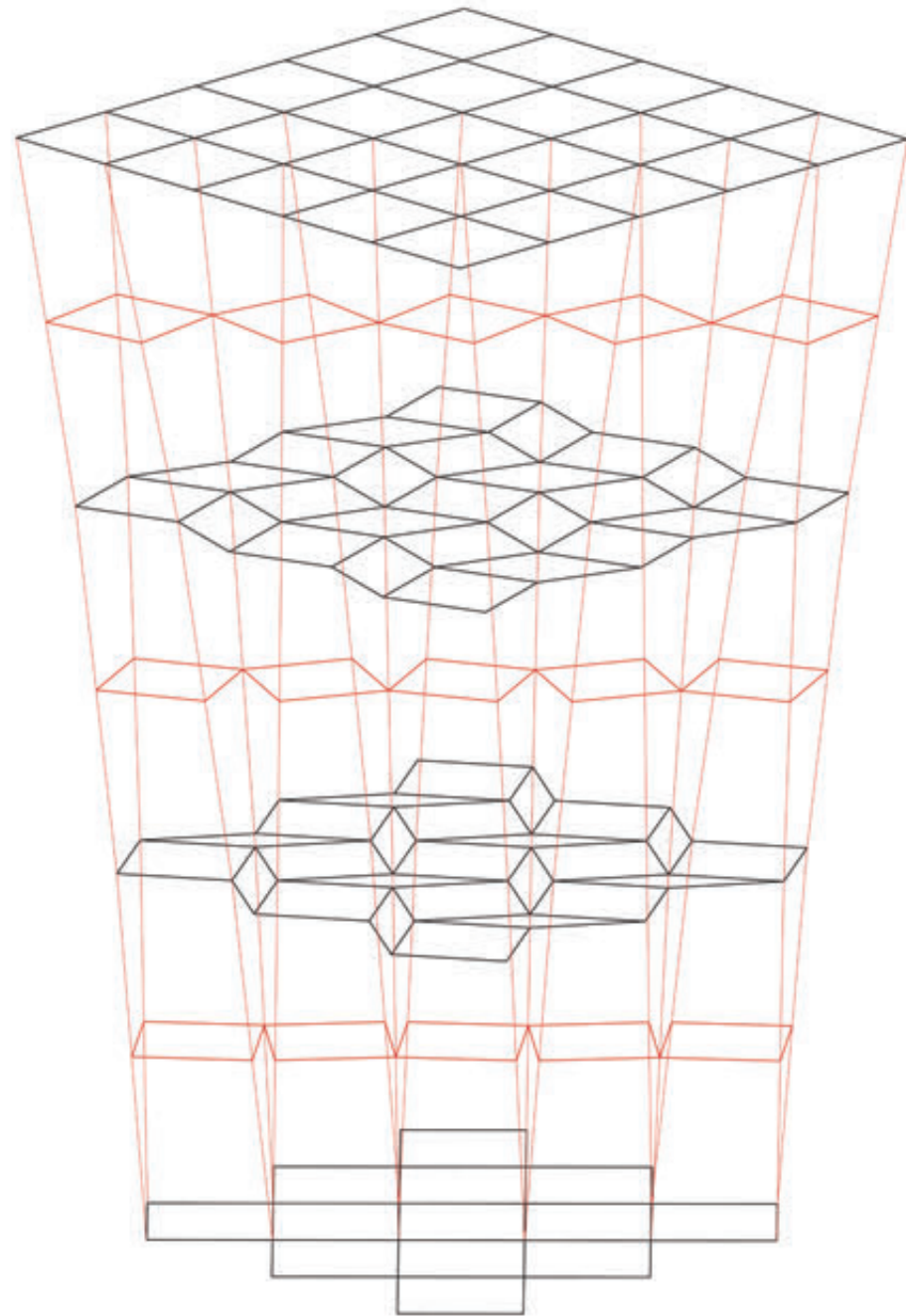




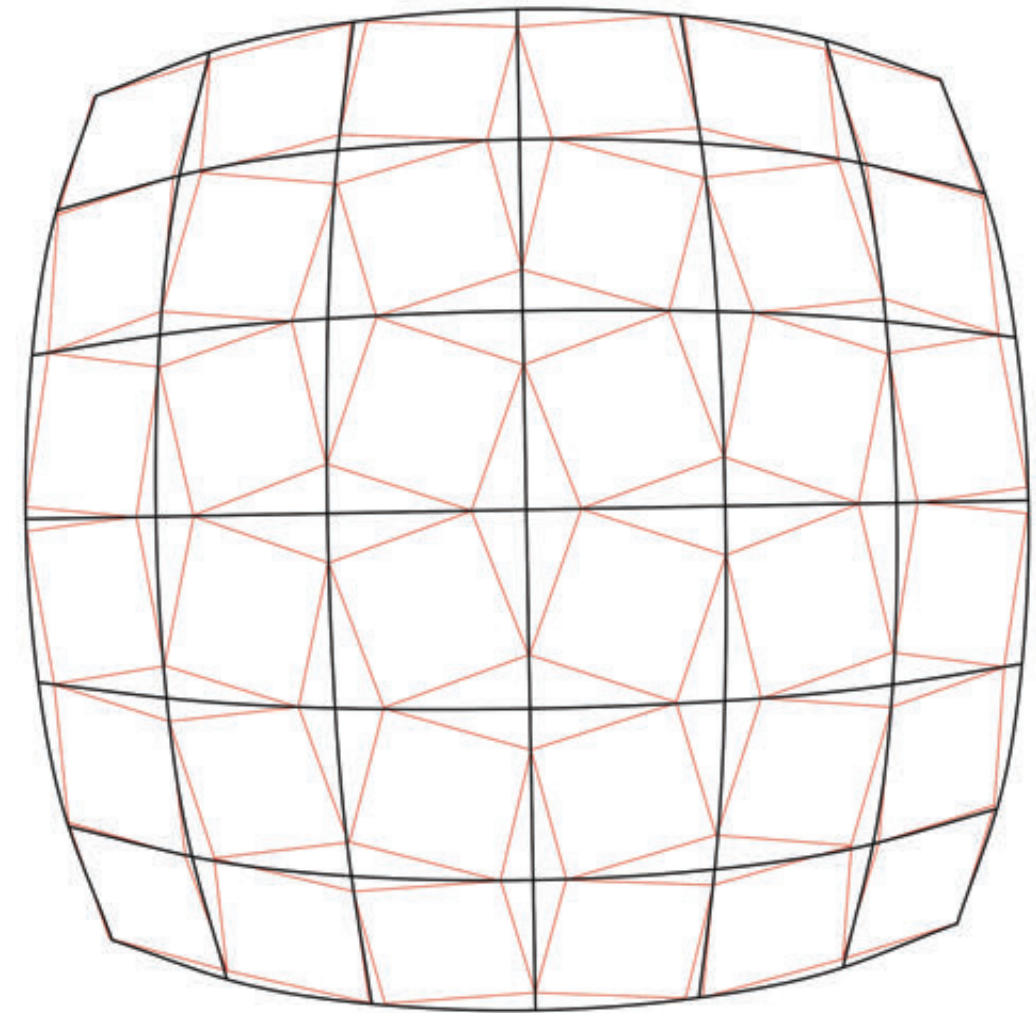


$$\begin{array}{ccccccc} & 0 & & 0 & & 0 & \\ & \downarrow & & \downarrow & & \downarrow & \\ 0 & \longrightarrow & A_1 & \longrightarrow & B_1 & \longrightarrow & C_1 \longrightarrow 0 \\ & \downarrow & & \downarrow & & \downarrow & \\ 0 & \longrightarrow & A_2 & \longrightarrow & B_2 & \longrightarrow & C_2 \longrightarrow 0 \\ & \downarrow & & \downarrow & & \downarrow & \\ 0 & \longrightarrow & A_3 & \longrightarrow & B_3 & \longrightarrow & C_3 \longrightarrow 0 \\ & \downarrow & & \downarrow & & \downarrow & \\ & 0 & & 0 & & 0 & \end{array}$$

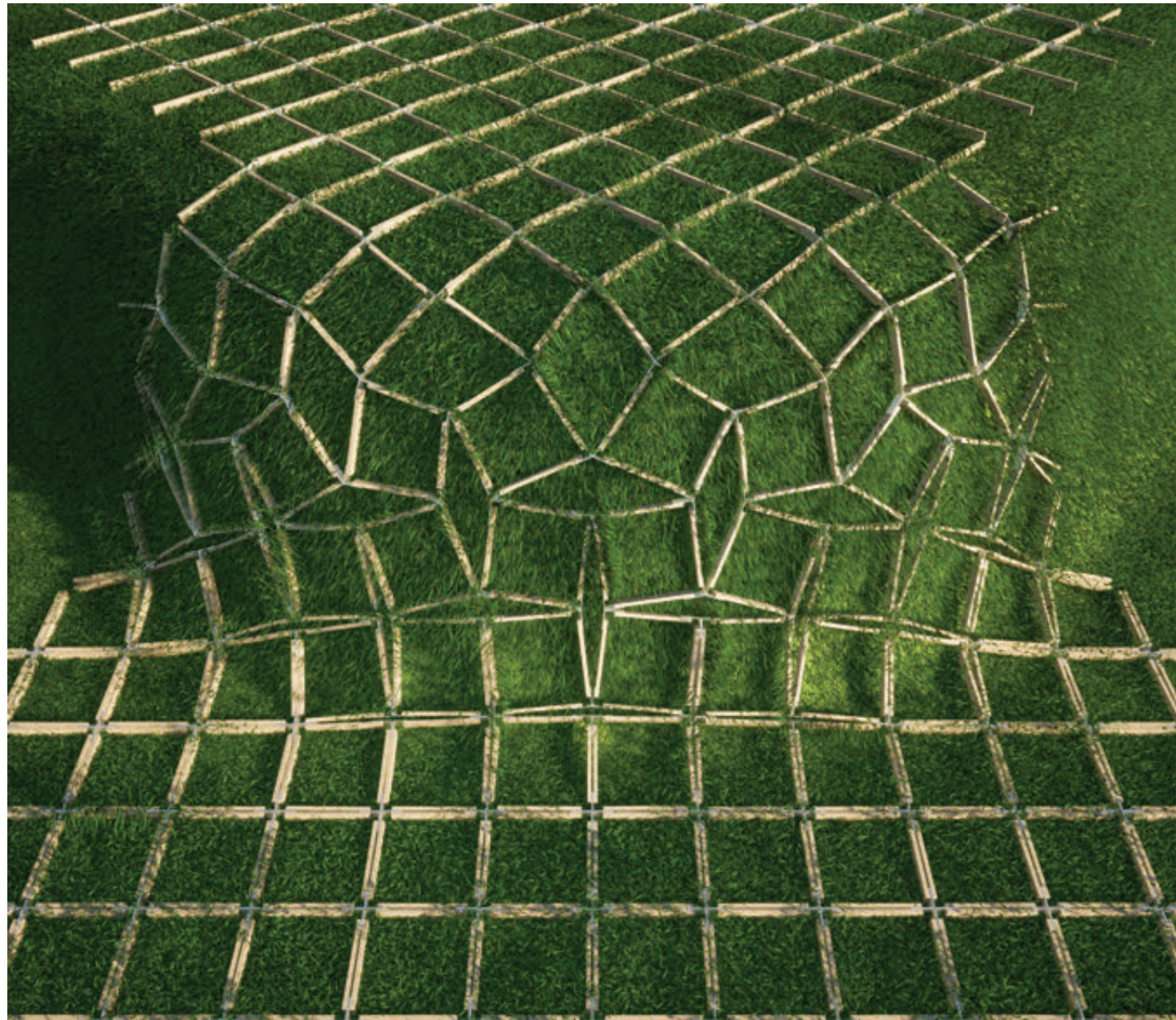
CONFORMAL AUXETICS



auxetics are assemblages of rigid members which expand uniformly in all directions when stretched



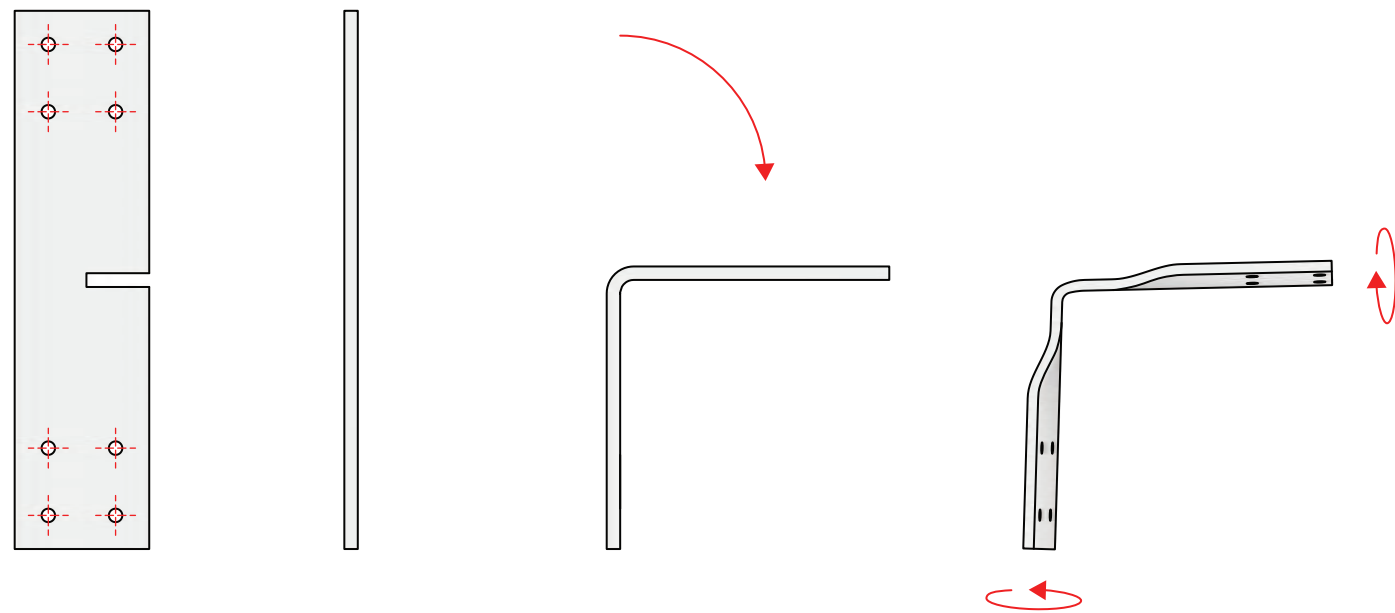
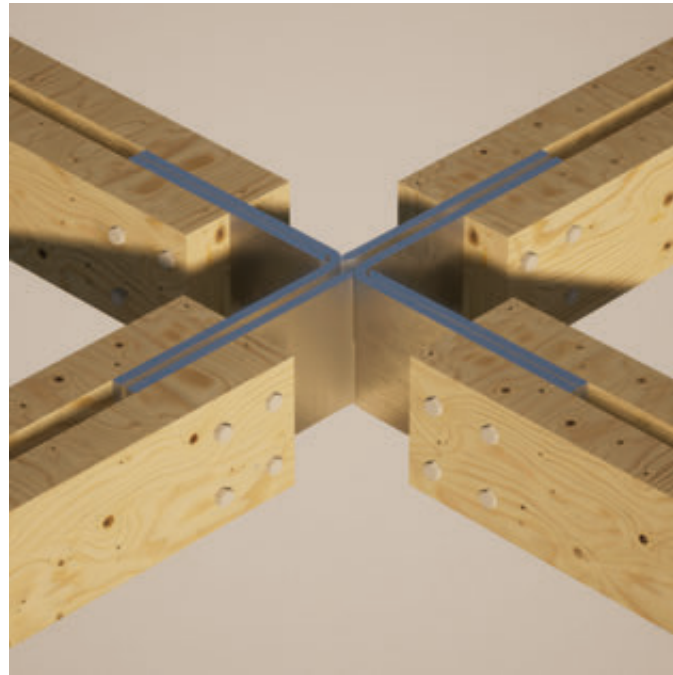
conformal mappings between surfaces are relationships which look locally like uniform expansion at each point



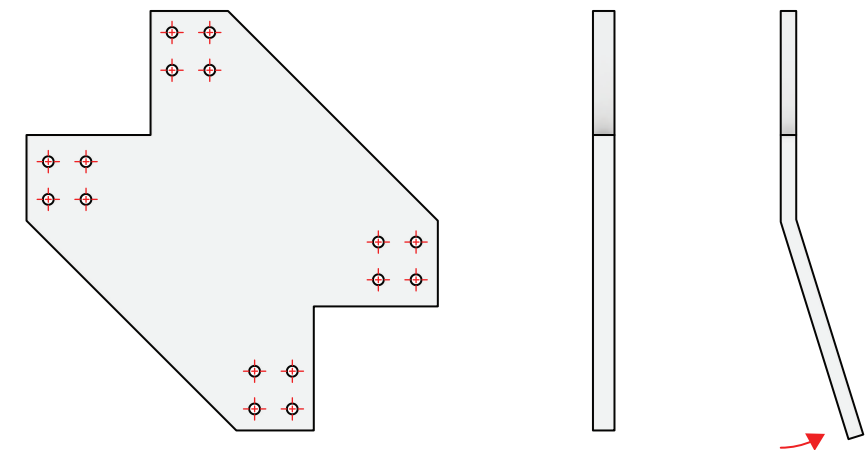
a conformal map between a **flat** and a **doubly-curved** surface provides a guide for an auxetic skeleton which reconciles a regular lattice with double curvature



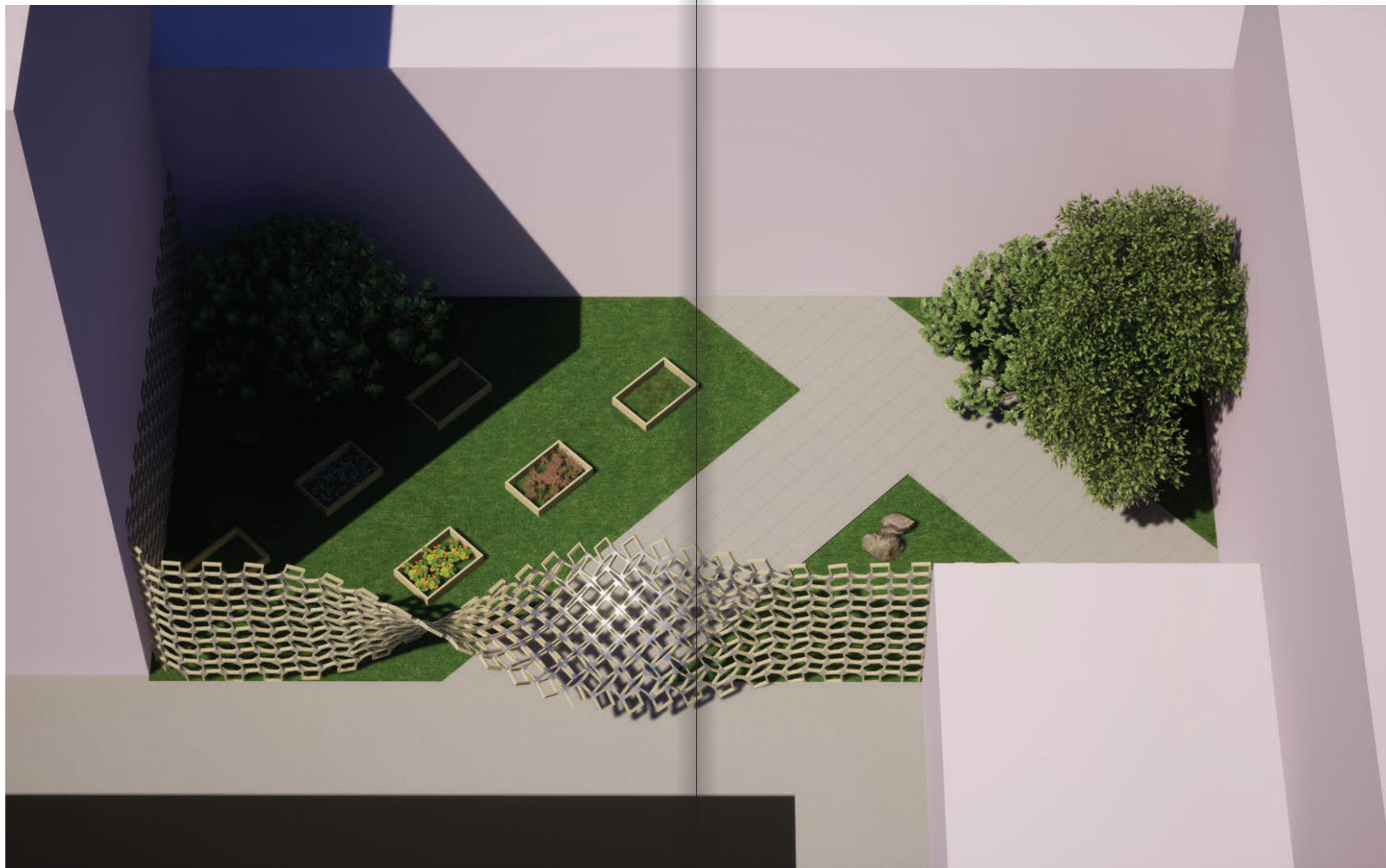
tuning the amount of curvature allows a square auxetic linkage to achieve both a **folded** and (dual) **unfolded** regular lattice

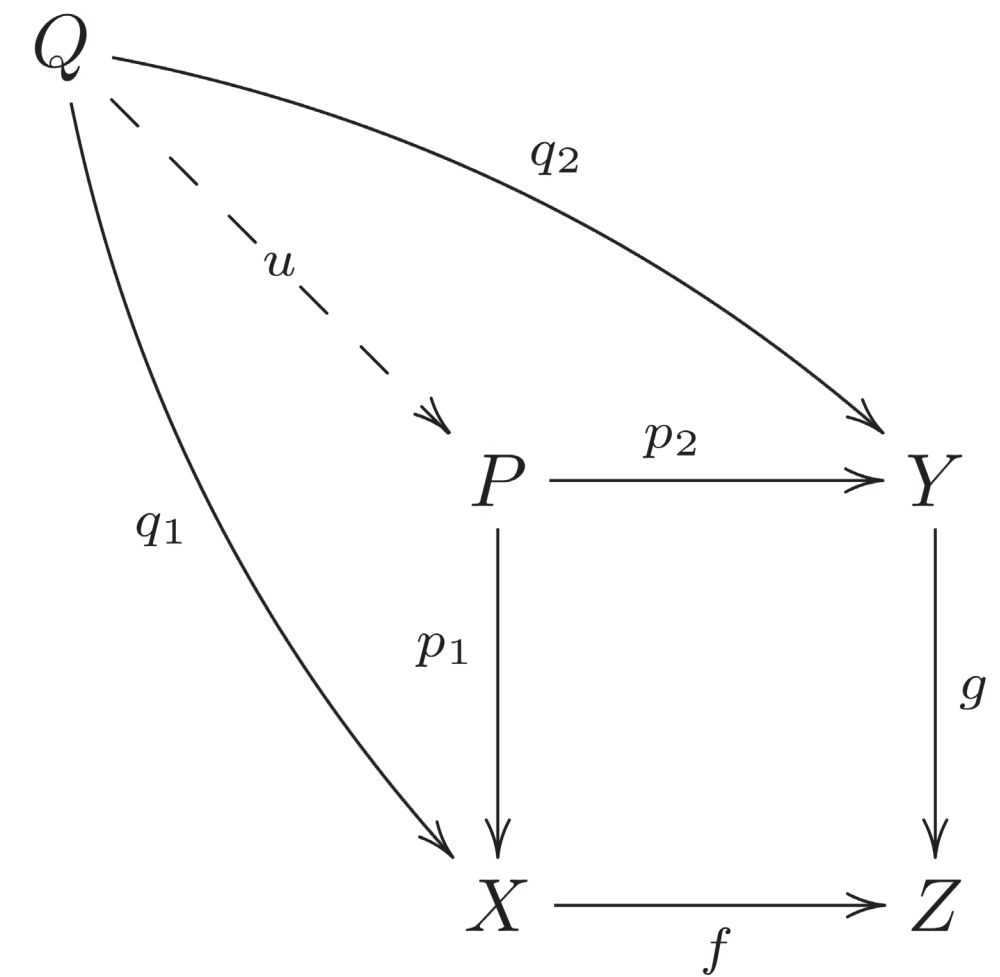


initial speculative universal joint for auxetic skeleton



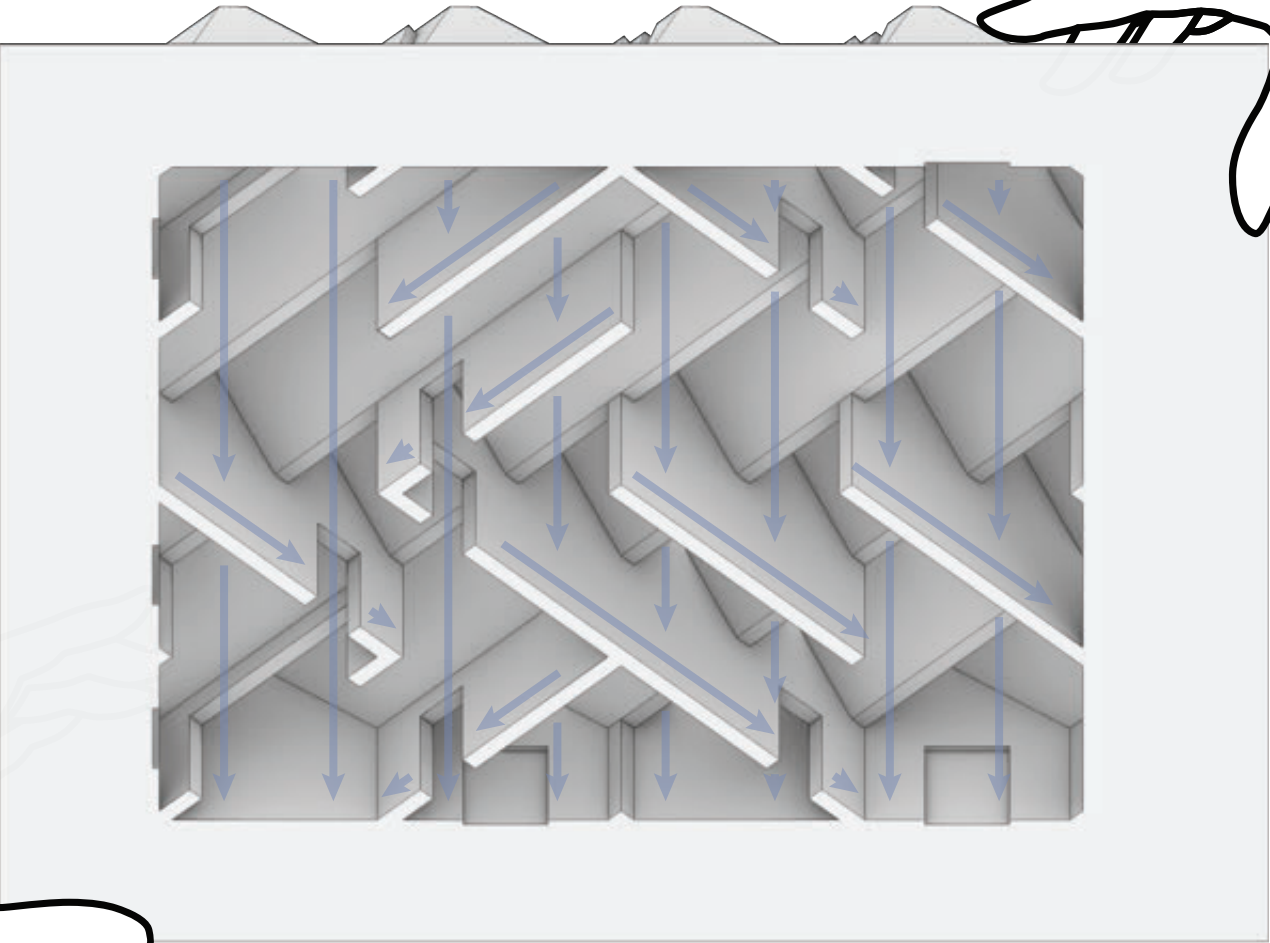
simplified speculative universal joint for auxetic skeleton



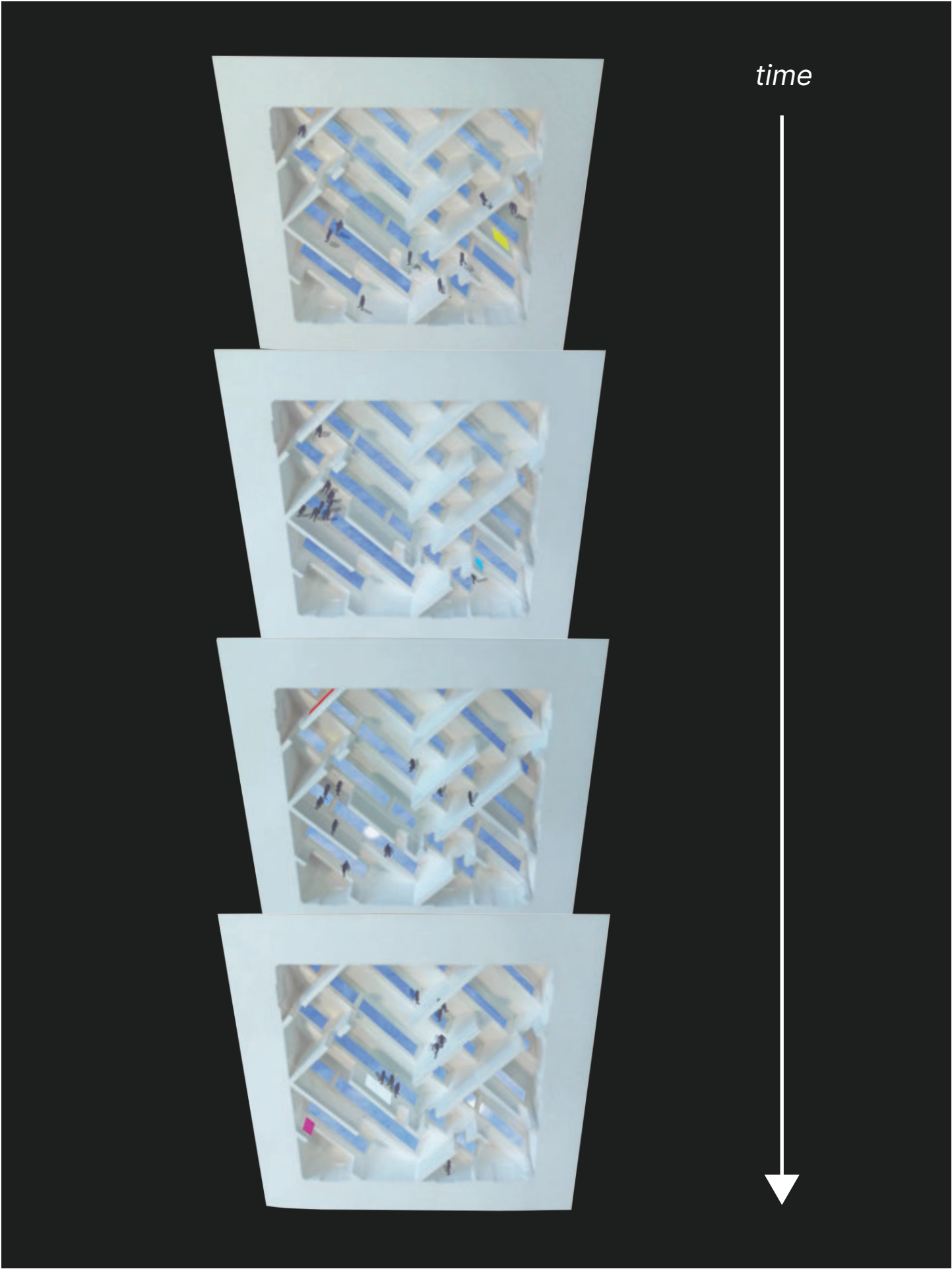


OTHER PLACES

Brief: Implement augmented reality layers of experience to existing architectural models as experiments in the design of intangible, **other** aspects of architecture.

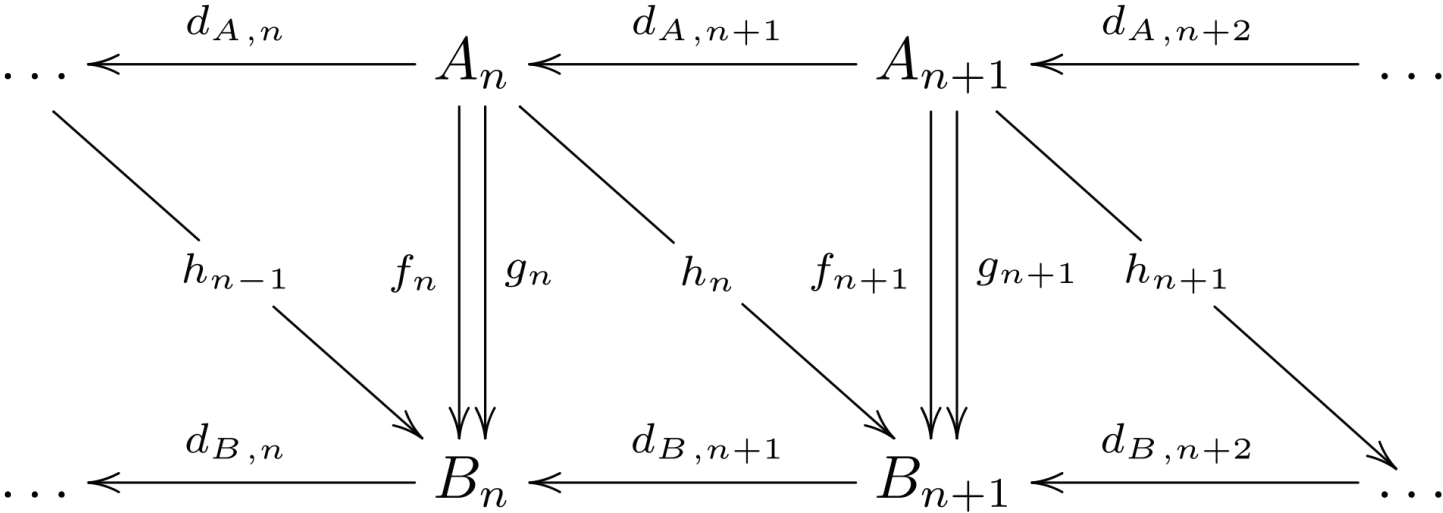
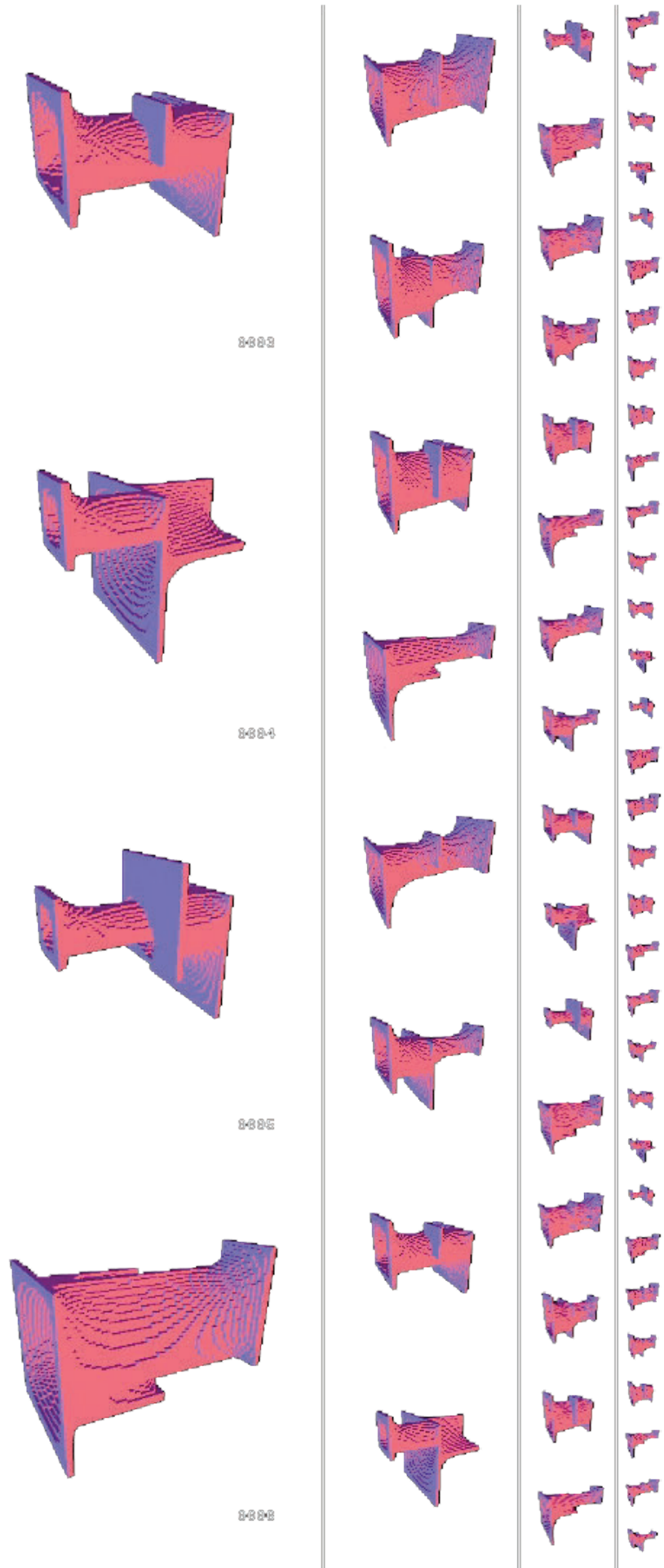
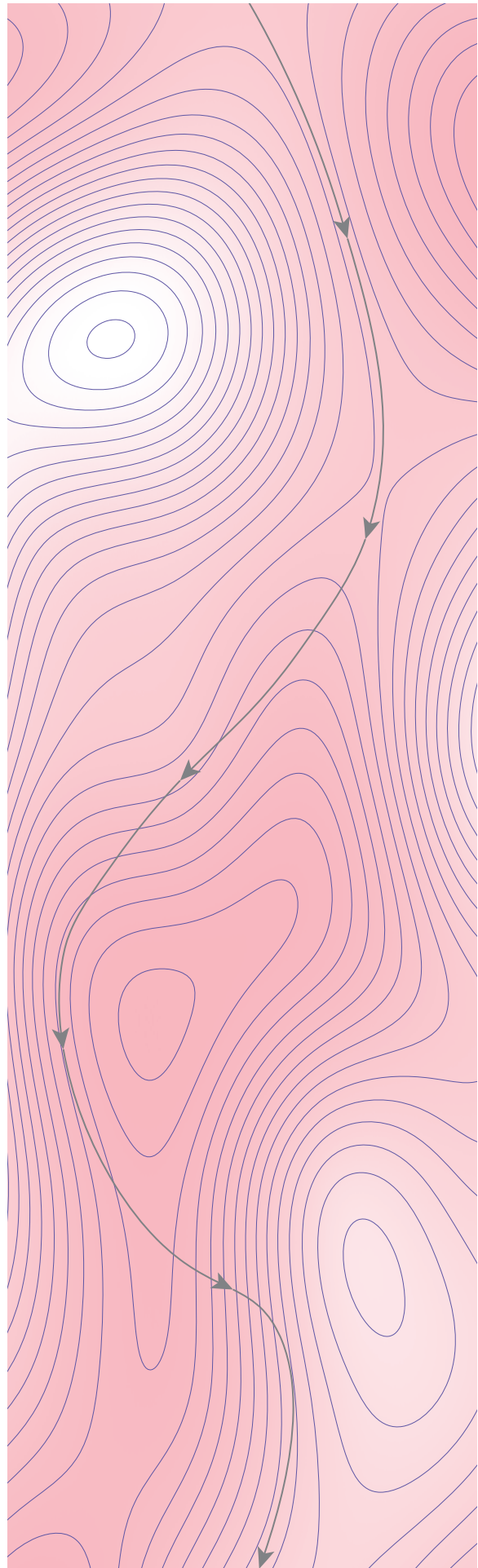


user manipulation of worm's eye model influences circulation of virtual agents navigating gallery throughout simulated day



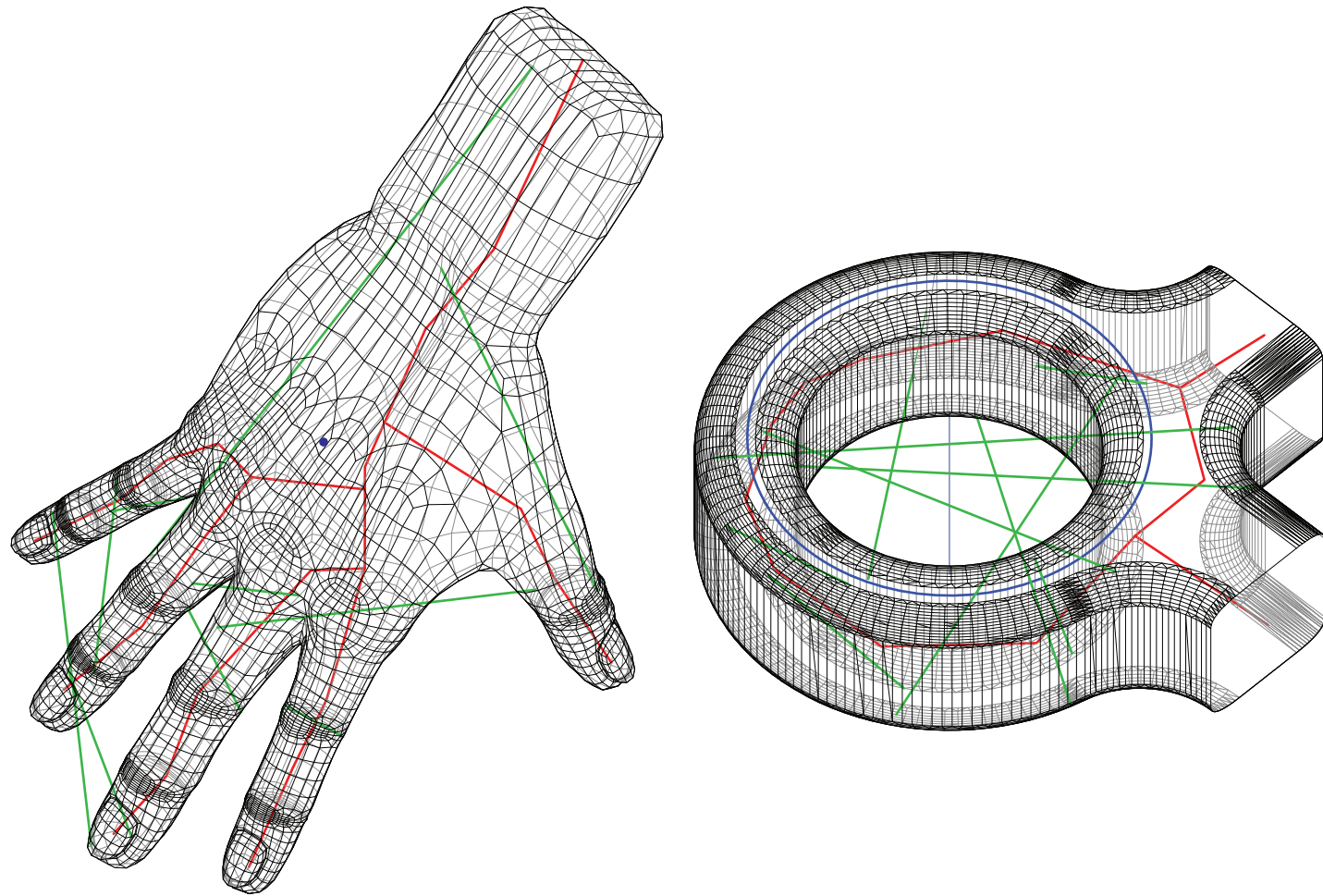


technical experiments in occlusion and avatar control as part of broader investigation into representation of privacy and agency in architectural models

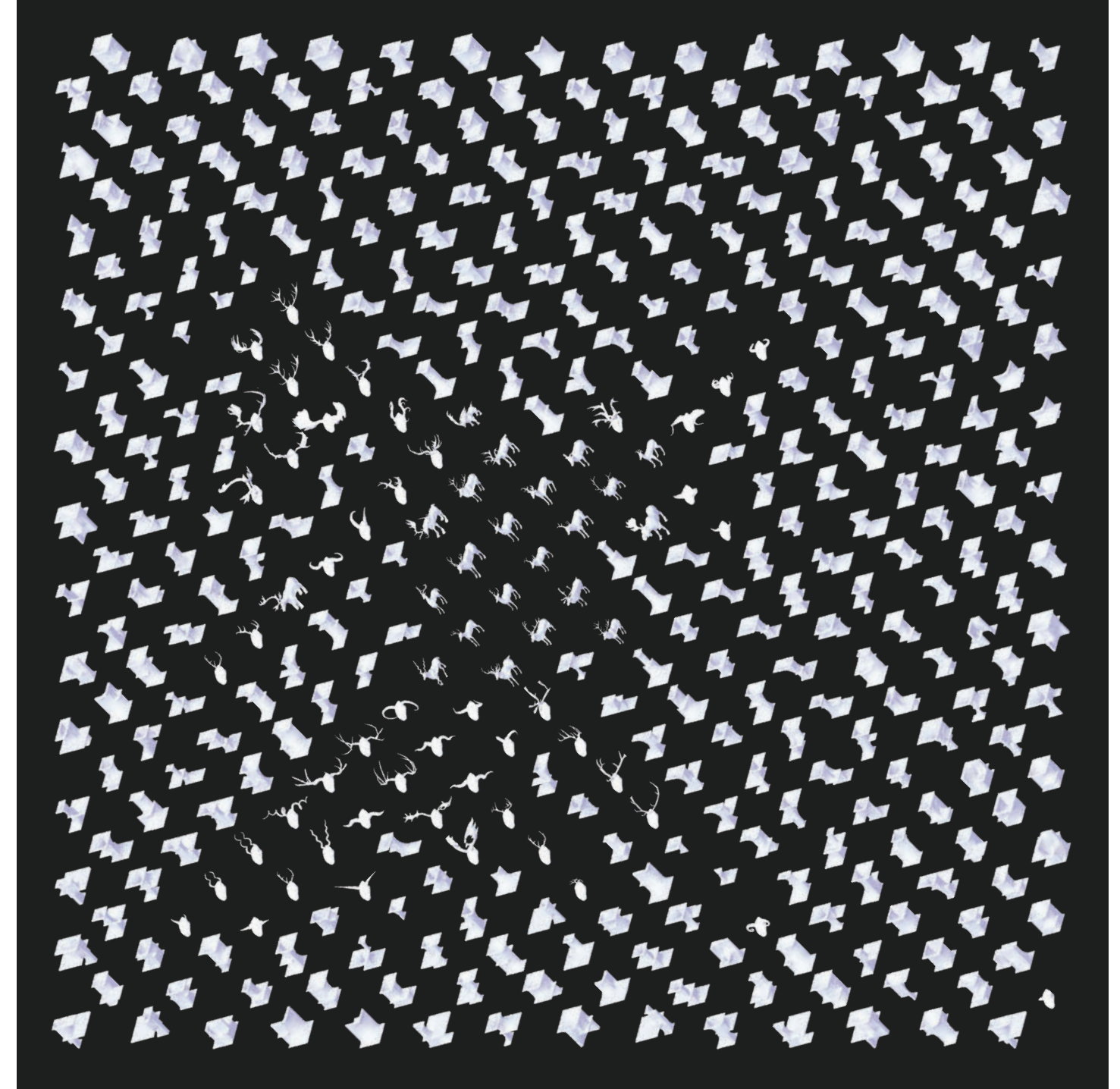


LATENT METRICS

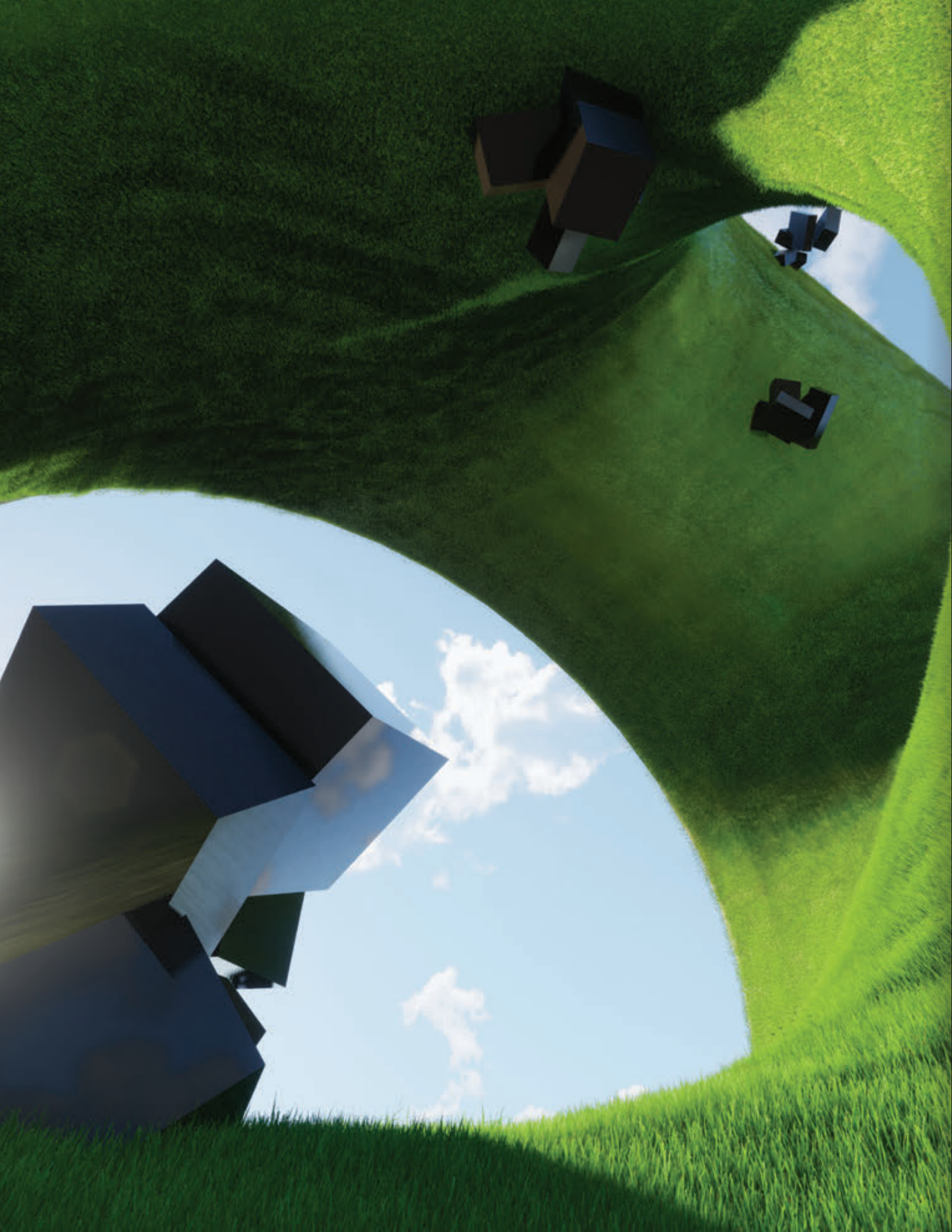
Brief: bias a 3D generative adversarial network (GAN) to capture intuitive notions of form so that the resultant high-dimensional **latent space** representation of 3D geometry has a more human-parsable topography



compiled metrics for comparing qualitative aspects of geometries:
skeletonization, **shape distribution**, **homology**



2D flattening of geometric latent space resulting in formal geography governed by
shape similarity



SPECULATIVE

PERSONAL PROJECT
2022

$$\begin{array}{ccc} Y & \xrightarrow{\tilde{f}_0} & E \\ \downarrow \iota_0 & \nearrow \tilde{f}_\bullet & \downarrow \pi \\ Y \times I & \xrightarrow{f_\bullet} & B \end{array}$$

FOLLY MATRIX

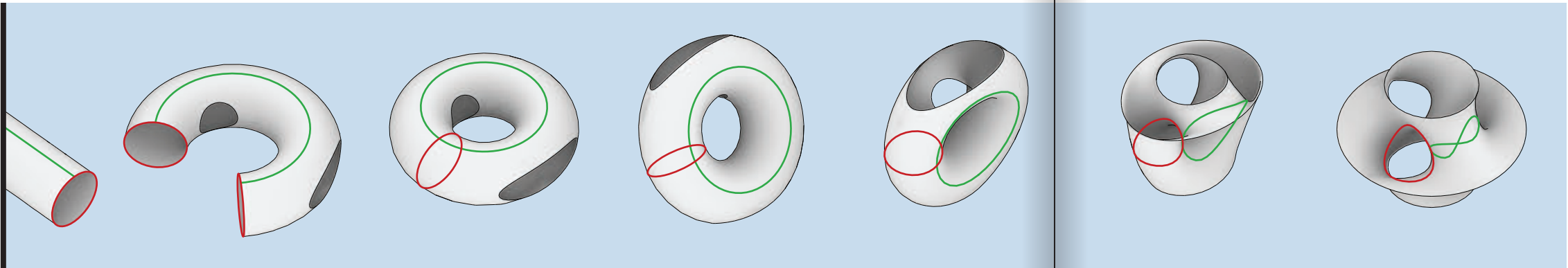
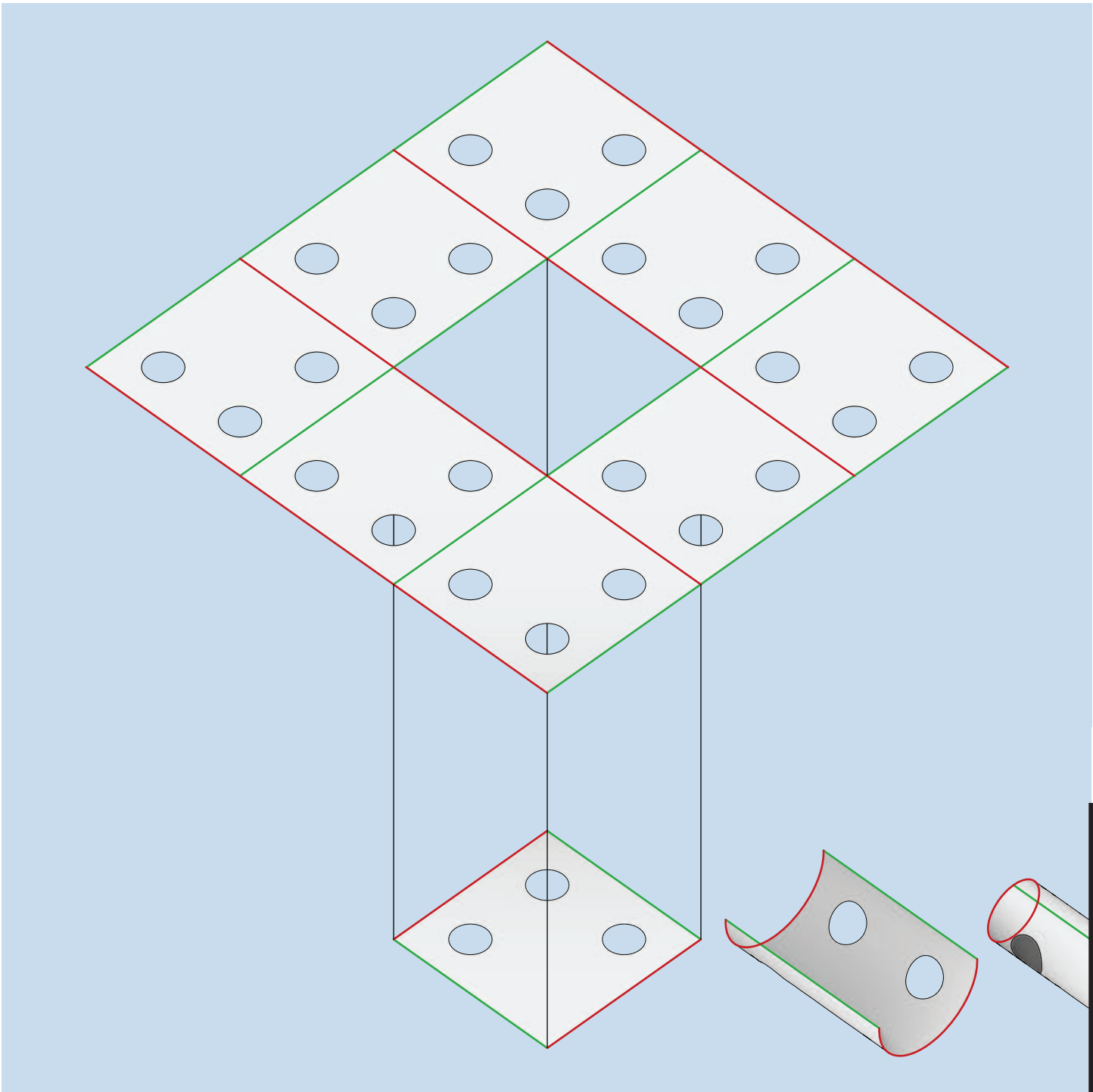
Brief: create a virtual gallery for architectural sketches and follies which one can get lost wandering in yet which is rigerously organized, one with dynamic spatial conditions yet is infinitely expandable

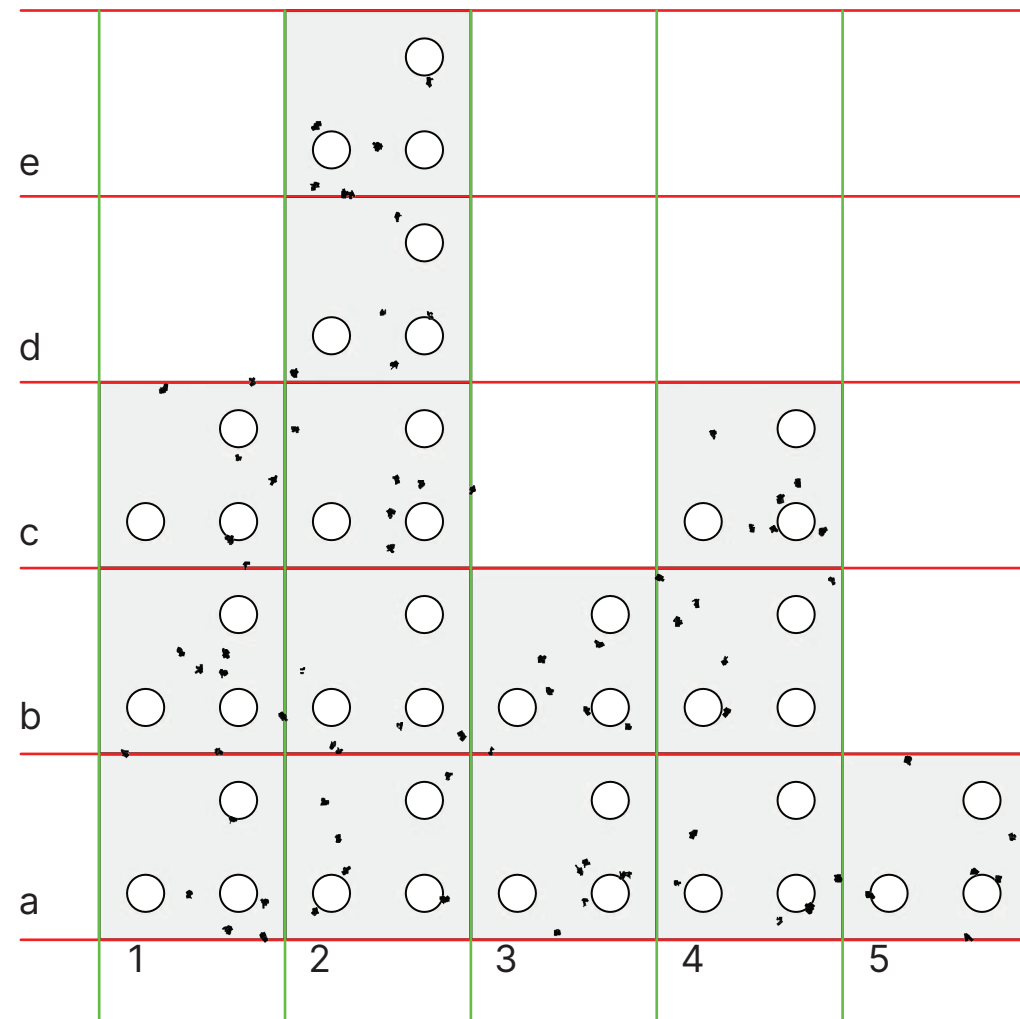
Definition 1.1 A **covering space** is a surface composed of copies of a given surface which are each mapped to that surface via a **covering map**. Covering spaces can be understood as an unfolding of the topological structure of the base surface.

Proposition 1.2 There exists a covering map from the infinite plane to the torus.

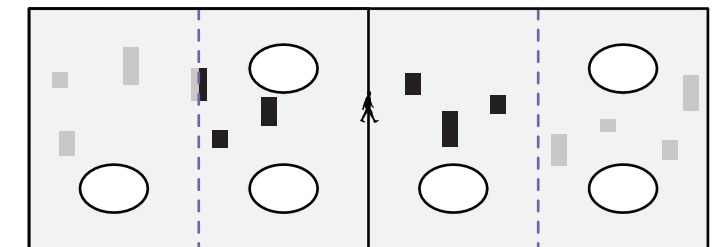
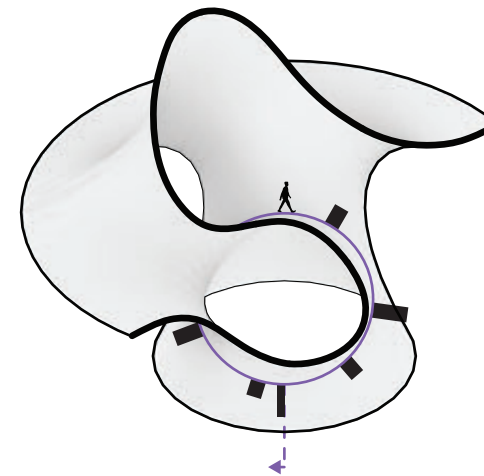
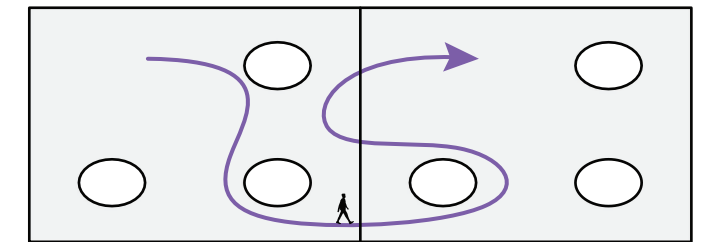
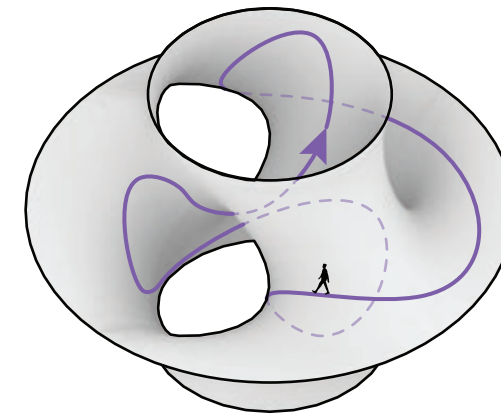
Proposition 1.3 A torus with three holes can be smoothly deformed into Costa's minimal surface.

Corollary 1.4 Composing these two propositions gives a covering map from the **infinite grid** with three holes per grid cell to **Costa's minimal surface**.

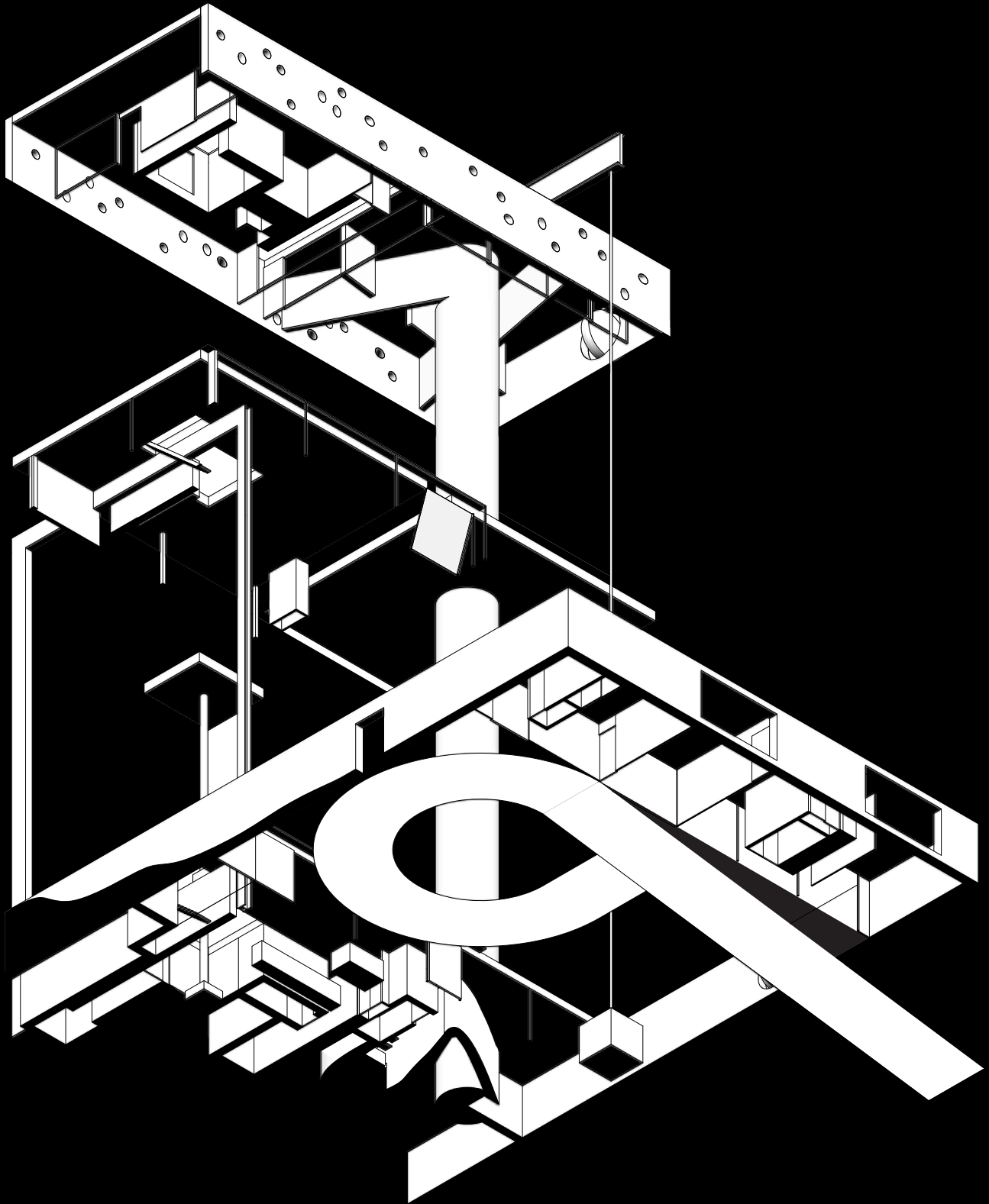




patrons can toggle between the gallery being experienced as a 3D minimal surface and a regular, infinite grid



due to the **Path Lifting Theorem**, Corollary 1.4 means that looping around a column of the minimal surface corresponds exactly to moving to an adjacent grid cell



*exploded axonometric drawing study of **Maison à Bordeaux** by **OMA***

